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Relative entropy based uncertainty principles for graph signals

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ABSTRACT

In physical quantum mechanics, the uncertainty principle in presence of quantum memory [Berta M, Christandl M, Colbeck R,et al., Nature Physics] can reach much lower bound, which has resulted in a huge breakthrough in quantum mechanics. Inspired by this idea, this paper would propose some novel uncertainty relations in terms of relative entropy for signal representation and time-frequency resolution analysis. On one hand, the relative entropy measures the distinguishability between the known (priori) basis and the client basis, which implies that we have partial "memory" of the client basis so that the uncertainty bounds become sharper in some cases. On the other hand, in some cases, if the reference basis along with nearly the same energy distribution could be given, then the uncertainty bound would tend to zero, as shows that there is no uncertainty any longer. These novel uncertainty relationships with sharper bounds would give us the potential advantages over the classical counterpart. In addition, the detailed comparison with classical Shannon entropy based uncertainty principle has been addressed as well via combined uncertainty relations. Finally, the theoretical analysis and numerical experiments on certain application over graph signals have been demonstrated to show the efficiency of these proposed relations.

1. Introduction

In 1948, Shannon proposed the concept of "information entropy" to solve the problem of quantifying and measuring information [1]. The amount of information in a piece of information is directly related to its uncertainty. For instance, to understand a very uncertain thing or something we know nothing about, we need to know a lot of information. On the contrary, if we already have a good understanding of something, we don't need too much information to determine it. Thus, from this perspective, we could believe that the measurement of information quantity is equal to the amount of uncertainty. On the other hand, more generally we often use uncertainty principle to measure the amount of uncertainty, first introduced by Heisenberg [2] and later various versions come into being such as Heisenberg uncertainty principles [3-10], Shannon entropic uncertainty principles [11-17] and windowed uncertainty principles [18] and so forth, which have been used widely in various fields such as physics, mathematics and information science and so forth.

For example, in the development of compressive sensing and signal sparse representation [18-23], the uncertainty principles on signal sparse representation have been introduced for signal representation and reconstruction. The milestone in this aspect is the work in [24] in 1989 and the work in [25] in 2001 by Donoho. Soon, one link between the uncertainty relations and the sparse signal reconstruction [26-28] were established by Donoho, Candès, Elad and et al. including the more general pairs of bases in terms of l_0-norm and l_1-norm.

On the other hand, with the development of graph signal processing, the graph uncertainty principles have been addressed in [31-34] and so forth by defining a "measure" of the signal in the vertex domain and the graph Fourier transform (GFT) domain. For instance, the study in [31] involves the unweighted graphs for "spreads" (i.e., vertex spread and spectral spread), which has been extended to weighted graphs in the work [35]. Differently, the work in [32] pays more attentions on an alternative definition and they focuses on the amount of energy to address the uncertainty principles over graph signals. Similarly, the study in [31,34] takes care of the smoothness of the signal in both domains using the graph difference operator. Furthermore, the work in [34] takes a non-local perspective where vertex and spectral measures are defined via various norms. Based on the fact that the identity tensor and the graph Fourier basis (GFB) form a pair of bases to represent graph signals, Perraudin et al. in [34] proposed the uncertainty relations in terms of graph signal sparseness, which shows that the "graph Fourier

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coherence" is a fundamental quantity for the uncertainty relations. At the same time, Teke and Vaidyanathan have proposed a new manner to formulate the graph uncertainty principles via a nonlocal measure in terms of sparseness in [38] by l_0-norm and l_1-norm and so forth. Besides the study [31-37], in [38] the work has promoted graph uncertainty principles to one new level. Recently, Xu et al. extended the Shannon entropic uncertainty relation and Rényi entropic uncertainty relation [39,40] on traditional vector signal sparse representation. However, there has been no any paper addressing graph signals on relative entropy based uncertainty relations up till now to our best knowledge [2-40].

In this paper, the relative entropy (i.e., relative Shannon entropy, which is also called as the Kullback-Leibler divergence that will be shown in the second section) based novel uncertainty relations on graph signals in terms of sparse representation would be deduced in great details. In physical quantum mechanics, it is well-known that the uncertainty principle in terms of quantum memory can reach much lower bound with a huge breakthrough in quantum mechanics [41]. Inspired by the idea of that in [41], this paper would propose some novel entropic uncertainty relations in terms of relative entropy that will have partial "memory" of the client basis so that the uncertainty bounds become sharper in some cases in information processing for signal representation. For the first time we will disclose the new relation that these new uncertainty bounds are not only related with the vector length, but also related with the min non-zero correlation between graph Fourier basis and reference graph Fourier basis, the max correlation between the two given graph Fourier basis and even the eigenvalues along with eigenvectors in graph matrix. Moreover, the relationship between the uncertainty bounds and the graph matrix eigenvalues would be discussed as well, as shows the new interesting interpretation on graph signal representation. In order to demonstrate the efficiency of these newly proposed uncertainty principles over graph signals, the theoretical analysis and numerical experiments would be shown in great details.

In the rest of this paper, Section 2 will show relative entropy for graph signals. In the third section the novel relative entropy based uncertainty relations for the graph vector signals are shown. Section 4 shows the graph matrix signals in terms of relative entropy based uncertainty principle for the first time, and the analysis will be shown as well. Section 5 shows the combined uncertainty principles. Section 6 will give the numerical examples to show the relations between bounds. Finally, the last section will conclude the work of this paper.

2. Relative entropy over graph

As shown in [41] by Berta and et al., the uncertainties in physical quantum mechanics about any quantity are always measured relative to the knowledge of an observer. That is to say, an observer with little knowledge will have high uncertainty while one with more knowledge will have less uncertainty. Inspired by this idea, in information science the concentration of signal would be in the similar manner if we adopt the similar measures as that in physics.

However, in classical signal processing there has been no same strategy as that in physical quantum mechanics in the form of presence of quantum memory. Fortunately, there is one measure, relative entropy, that could be used to measure the uncertainty at the same time with more priori knowledge. The relative entropy, called as Kullback–Leibler divergence, also known as Kullback-Leibler divergence or information divergence, is an asymmetric measure of the difference between two probability (or signal) distributions. In information theory, Kullback–Leibler divergence is equivalent to the difference between the information entropies of two probability distributions [42]. Kullback–Leibler divergence is the loss function of some optimization algorithms, such as the Expectation–maximization algorithm (EM) [43]. At this time, one probability distribution involved in the calculation is the real distribution, and the other is the theoretical (fitting) distribution. Kullback–Leibler divergence represents the information loss generated when the theoretical distribution is used to fit the real distribution. That is to say, if we could give the expectant (assumed) concentration of signal, and the relative entropy would imply the difference between our given expectant (assumed) signal concentration and the practical signal concentration (or signal distribution). Very specially, if the given expectant (assumed) signal concentration and the practical signal concentration are the same, then the relative entropy will be zero, at this time, the uncertainty would be zero. Instead, if the given expectant (assumed) signal concentration and the practical signal concentration are fully different, then the relative entropy will be very high, at this time, the uncertainty would be very high. That is to say, if we employ the relative entropy to measure our signal concentration under the case of given expectant (assumed) signal distributions, then the uncertainty will reduce in some cases that the given expectant (assumed) signal concentration and the practical signal concentration are overlapped. Thus, the idea of [41] by Berta and et al. would be introduced into our signal concentration analysis such as sparse representation and so forth.

On the other hand, the graph theory, as an important branch of mathematical science and information theory, has gradually proved to be one important scientific research field [44] because a lot of data naturally belong to graph signals. For example, with the rising of internet networks, the relations in the virtual world have become a bridge for information dissemination and the information of individual internet nodes has become the discrete grid data [45,46]. Moreover, the traditional regular signal can also be treated as the special case of graph signal, e.g., establishing a similarity map [47-49] according to the similarity between data can realize some functions in machine learning. In other words, with the rapid development of information technology, the emergence of a large number of multidimensional, distributed and irregular discrete grid data has resulted in new challenges and more complicated processing requirements [50,51] so that the processing of these data involves the emerging field: graph signal processing (GSP).

A graph $\mathbf{G}$ (e.g., see Fig. 1(a)) is defined by the set of vertices $\boldsymbol{v}_1$ (see $\boldsymbol{v}_1 = \{0, 1, 2, 3\}$in Fig. 1(a)) and the set of edges Γ_1 whose connected weights are shown in weight matrix $\mathbf{A}_1$ (e.g., see the blue lines in Fig. 1 (a)) connecting the vertices. A graph can be undirected and directed. In Fig. 1(b), there is another graph defined by the set of $\boldsymbol{v}_2$ ($\boldsymbol{v}_2 = \{0,1,2,3\}$) and edges Γ_2 whose connected weights are shown in weight matrix $\mathbf{A}_2$ with the same number of vertices as that of Fig. 1(a). The difference between them is the edges. Clearly, Fig. 1(b) is a circle graph (according to the traditional 1D signal). In Fig. 1, the two graphs are undirected. Therefore, we could obtain their adjacent matrices as follows respectively:

$$\mathbf{A}_1 = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix} \text{and}\, \mathbf{A}_2 = \begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}. \tag{1}$$

Clearly, the two graphs are similar, which can be found by the two adjacent matrices in (1). The two adjacent matrices are real symmetric here. In this paper, we only discuss the real symmetric adjacent matrix

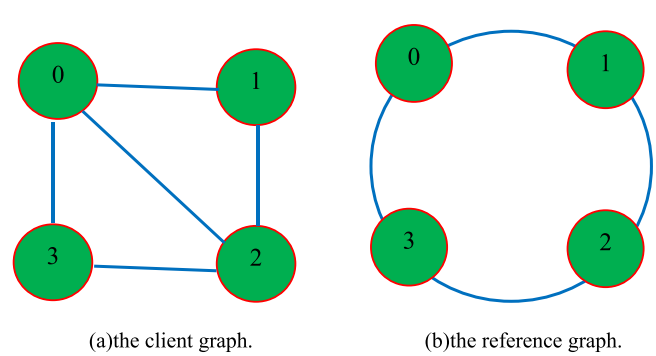


Fig. 1. One simple example of two graphs.

like that shown in Eq. (1).

In Fig. 1, if we know the data at all the vertices on time t, then we have the graph signals as follows:

$$\mathbf{v}_1 = \left[v_{1,1}(t)\ v_{1,2}(t)\ v_{1,3}(t)\ v_{1,4}(t)\right]^T \text{and}\, \mathbf{v}_2 = \left[v_{2,1}(t)\ v_{2,2}(t)\ v_{2,3}(t)\ v_{2,4}(t)\right]^T. \tag{2}$$

On the other hand, we also can obtain the Laplace matrices of these two graphs by subtracting their degree matrices from adjacent matrices as follows:

$$\mathbf{L}_1 = \mathbf{D}_1 - \mathbf{A}_1 = \begin{bmatrix} 3 & -1 & -1 & -1 \\ -1 & 2 & -1 & 0 \\ -1 & -1 & 3 & -1 \\ -1 & 0 & -1 & 2 \end{bmatrix}, \tag{3–1}$$

$$\mathbf{L}_2 = \mathbf{D}_2 - \mathbf{A}_2 = \begin{bmatrix} 2 & -1 & 0 & -1 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ -1 & 0 & -1 & 2 \end{bmatrix}. \tag{3–2}$$

At the same time, we can obtain their eigendecomposition as follows

$$\mathbf{L}_1 = \mathbf{U}_1\boldsymbol{\Lambda}_1\mathbf{U}_1^T,\ \mathbf{L}_2 = \mathbf{U}_2\boldsymbol{\Lambda}_2\mathbf{U}_2^T, \tag{4}$$

where $\mathbf{U}_1^T$ is the graph Fourier basis of graph in Fig. 1(a) and $\mathbf{U}_2^T$ is the graph Fourier basis of graph in Fig. 1(b).

In some cases, if the two graphs are similar (i.e., the weight value between nodes 0 and 2 of Fig. 1(a) is nearly zero instead of weight 1 and the other weight values in Fig. 1(a) are the same as that in Fig. 1(b)), it is possible that their eigendecomposition is also similar (note the word "similar", but not the word "same"). At this time, we can take one graph as the reference graph, i.e., the given graph basis in advance by *a priori* can be taken as the reference basis. Therefore, we can obtain the relative entropy related with basis $\mathbf{U}_1$ and $\mathbf{U}_2$ as follows:

$$\mathbf{H}^{\mathbf{U}_1\|\mathbf{U}_2}(\mathbf{v}_1) = \sum\nolimits_i^n \left|\mathbf{y}_i^\dagger\mathbf{v}_1\right|^2 \log\frac{\left|\mathbf{y}_i^\dagger\mathbf{v}_1\right|^2}{\left|\mathbf{z}_i^\dagger\mathbf{v}_1\right|^2} \text{while}\, \mathbf{U}_1 = [\mathbf{y}_1\ \mathbf{y}_2\ \cdots\ \mathbf{y}_n]\, \text{and}\, \mathbf{U}_2 = [\mathbf{z}_1\ \mathbf{z}_2\ \cdots\ \mathbf{z}_n]. \tag{5}$$

Here in this paper we set $\log = \log_2$ and $\ln = \log_e$, $\|\mathbf{v}_1\|_2^2 = 1$ without loss of generality. $\mathbf{U}_1(\mathbf{U}_1^T)$ is the given client graph Fourier basis and $\mathbf{U}_2(\mathbf{U}_2^T)$ is the given reference graph Fourier basis, and they belong to unitary matrix. $\mathbf{v}_1$ is the given graph vector signal to be represented or processed. $\mathbf{y}_i^\dagger\mathbf{v}_1$ denotes how much $\mathbf{y}_i$ (from the client graph Fourier basis $\mathbf{U}_1$) in $\mathbf{v}_1$, and $\mathbf{z}_i^\dagger\mathbf{v}_1$ denotes how much $\mathbf{z}_i$ (from the client graph Fourier basis $\mathbf{U}_2$) in $\mathbf{v}_1$. Therefore, we have $\sum_i^n \left|\mathbf{y}_i^\dagger\mathbf{v}_1\right|^2 = 1$. Clearly, if $\mathbf{U}_1 = \mathbf{U}_2$, then $\mathbf{H}^{\mathbf{U}_1\|\mathbf{U}_2}(\mathbf{v}_1) = 0$. That is to say, the more overlapping parts $\mathbf{U}_1$ and $\mathbf{U}_2$ have (more similar they are), then in principle the less $\mathbf{H}^{\mathbf{U}_1\|\mathbf{U}_2}(\mathbf{v}_1)$ will be, and vice versa.

Generally, we often ignore the variable t to have the graph signals $\mathbf{v}_1 = \left[v_{1,1}\ v_{1,2}\ v_{1,3}\ v_{1,4}\right]^T$ and $\mathbf{v}_2 = \left[v_{2,1}\ v_{2,2}\ v_{2,3}\ v_{2,4}\right]^T$ if we needn't consider the time variable t. However, if we have multiple time instants for every vertex, then at every vertex there will be multiple data shown as follows:

$$\mathbf{V}_1 = \begin{bmatrix} v_{1,1}(t_1)\ v_{1,2}(t_1)\ v_{1,3}(t_1)\ v_{1,4}(t_1) \\ v_{1,1}(t_2)\ v_{1,2}(t_2)\ v_{1,3}(t_2)\ v_{1,4}(t_2) \\ \cdots\cdots\cdots\cdots \\ v_{1,1}(t_n)\ v_{1,2}(t_n)\ v_{1,3}(t_n)\ v_{1,4}(t_n) \end{bmatrix}^T \text{and}\, \mathbf{V}_2 = \begin{bmatrix} v_{2,1}(t_1)\ v_{2,2}(t_1)\ v_{2,3}(t_1)\ v_{2,4}(t_1) \\ v_{2,1}(t_2)\ v_{2,2}(t_2)\ v_{2,3}(t_2)\ v_{2,4}(t_2) \\ \cdots\cdots\cdots\cdots \\ v_{2,1}(t_n)\ v_{2,2}(t_n)\ v_{2,3}(t_n)\ v_{2,4}(t_n) \end{bmatrix}^T. \tag{6}$$

See Fig. 2, we find that with time going, we can obtain the Cartesian product of two graphs. Here the top graphs are the results of the Cartesian product between the bottom graph in Fig. 2 and the two graphs in Fig. 1. The bottom graph in Fig. 2 is only a line graph with n time instants (i.e., a line graph with n vertices and here we let $n = 4$).

In order to simplify our analysis, we assume our graph signals are square matrices because we can make them square via replenishing zeros or segmenting. At the same time, in many cases, we needn't care the time variables and we only take care of the graph data, so we obtain the following graph matrix signals

$$\mathbf{V}_1 = \begin{bmatrix} v_{1,11}\ v_{1,12}\ v_{1,13}\ v_{1,14} \\ v_{1,21}\ v_{1,22}\ v_{1,23}\ v_{1,24} \\ \cdots\cdots\cdots \\ v_{1,n1}\ v_{1,n2}\ v_{1,n3}\ v_{1,n4} \end{bmatrix}^T \text{and}\, \mathbf{V}_2 = \begin{bmatrix} v_{2,11}\ v_{2,12}\ v_{2,13}\ v_{2,14} \\ v_{2,21}\ v_{2,22}\ v_{2,23}\ v_{2,24} \\ \cdots\cdots\cdots \\ v_{2,n1}\ v_{2,n2}\ v_{2,n3}\ v_{2,n4} \end{bmatrix}^T. \tag{7}$$

Now in some cases it is possible we assume that the graph matrix signal of $\mathbf{V}_1$ can be decomposed as the following relation

$\mathbf{V}_1 = \sum_k \lambda_k \mathbf{v}_{1k}\mathbf{v}_{1k}^\dagger$ and $\|\mathbf{v}_{1k}\|_2^2 = 1$ without loss of generality, which is indeed the matrix spectral decomposition (or matrix eigendecomposition). In practice, in many cases this decomposition will exist for both the graph signals and regular signals such as images [43,49].

Then we will have the following relative entropy related with basis $\mathbf{U}_1$ and $\mathbf{U}_2$

$$\mathbf{H}^{\mathbf{U}_1\|\mathbf{U}_2}(\mathbf{V}_1) = \sum\nolimits_{k=1}^{\mathrm{K}} \left(\lambda_k \sum\nolimits_{i-1}^n \left|\mathbf{x}_i^\dagger\mathbf{v}_{1k}\right|^2 \log\frac{\left|\mathbf{x}_i^\dagger\mathbf{v}_{1k}\right|^2}{\left|\mathbf{z}_i^\dagger\mathbf{v}_{1k}\right|^2}\right) \text{while}\, \mathbf{U}_1 = [\mathbf{x}_1\ \mathbf{x}_2\ \cdots\ \mathbf{x}_n]\, \text{and}\, \mathbf{U}_2 = [\mathbf{z}_1\ \mathbf{z}_2\ \cdots\ \mathbf{z}_n]. \tag{8}$$

$\mathbf{U}_1(\mathbf{U}_1^T)$ is the given client graph Fourier basis and $\mathbf{U}_2(\mathbf{U}_2^T)$ is the given reference graph Fourier basis, and they belong to unitary matrix. $\mathbf{V}_1$ is the given graph matrix signal to be represented or processed. $\mathbf{x}_i^\dagger\mathbf{v}_{1k}$ denotes how much $\mathbf{x}_i$ (from the client graph Fourier basis $\mathbf{U}_1$) in $\mathbf{v}_{1k}$

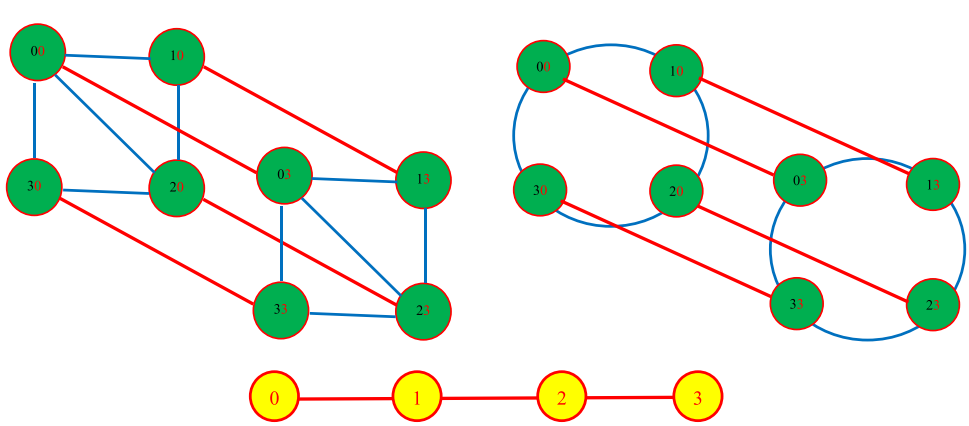


Fig. 2. The graphs produced by the Cartesian product (e.g., with time going).

(from the graph matrix signal of $\mathbf{V}_1$), and $\mathbf{z}_i^\dagger \mathbf{v}_{1k}$ denotes how much $\mathbf{z}_i$ (from the client graph Fourier basis $\mathbf{U}_2$) in $\mathbf{v}_{1k}$ (from the graph matrix signal of $\mathbf{V}_1$). Thus we have $\sum_{i-1}^n |\mathbf{x}_i^\dagger \mathbf{v}_{1k}|^2 = 1$. Clearly, if $\mathbf{U}_1 = \mathbf{U}_2$, then $\mathbf{H}^{\mathbf{U}_1\|\mathbf{U}_2}(\mathbf{v}_1) = 0$. That is to say, the more overlapping parts $\mathbf{U}_1$ and $\mathbf{U}_2$ have (more similar they are), then in principle the less $\mathbf{H}^{\mathbf{U}_1\|\mathbf{U}_2}(\mathbf{v}_1)$ will be, and vice versa. Specially, if $K = 1$, then $\mathbf{H}^{\mathbf{U}_1\|\mathbf{U}_2}(\mathbf{V}_1) = \mathbf{H}^{\mathbf{U}_1\|\mathbf{U}_2}(\mathbf{v}_1)$, which reduces to the case of vector in (5).

3. Relative entropy based uncertainty principles on graph vector

The scalar, vector and tensor are the main three classes of data in graph signal processing [26] and information science and so forth. In practice, we could say that the scalar is the zero-order tensor, and the vector is the first-order tensor. Instead, the matrix can be considered as the second-order tensor. Generally speaking, the matrices and higher-order tensors are often called as tensors. Since the scalar is only one single number, the uncertainty principle on scalar is trivial. Thus, in this paper we will pay more attentions on the graph vector and graph matrix in terms of uncertainty principle associated with reference basis via relative entropy.

Theorem 1. *Assume that we have three graphs ($\mathbf{G}_1$, $\mathbf{G}_2, \mathbf{G}_3$) and client graph $\mathbf{G}_1$ has some similar graph structure and graph signal with that of reference graph $\mathbf{G}_2, \mathbf{G}_3$. For the given client graph $\mathbf{G}_1$ with its vector signal $\mathbf{v} = [v_1\ v_2 \cdots v_n]^T$ satisfying $\mathrm{tr}\left((\mathbf{v}\mathbf{v}^\dagger)^2\right) = 1$, if we have four series of standard orthogonal graph basis $\mathbf{X} = [\mathbf{x}_1\ \mathbf{x}_2 \cdots \mathbf{x}_n], \mathbf{Y} = [\mathbf{y}_1\ \mathbf{y}_2 \cdots \mathbf{y}_n], \mathbf{Z}_1 = [\mathbf{z}_{1,1}\ \mathbf{z}_{1,2} \cdots \mathbf{z}_{1,n}]$, $\mathbf{Z}_2 = [\mathbf{z}_{2,1}\ \mathbf{z}_{2,2} \cdots \mathbf{z}_{2,n}]$ and two unitary transform matrix $\mathbf{U}_1, \mathbf{U}_2$ satisfying $\mathbf{X} = \mathbf{U}_1\mathbf{Y}, \mathbf{Y} = \mathbf{U}_2\mathbf{Z}$, $\mathbf{X}$ is the graph Fourier basis of the graph $\mathbf{G}_1$, and $\mathbf{Z}_1, \mathbf{Z}_2$is the reference Fourier basis of graph $\mathbf{G}_2$, $\mathbf{G}_3$, respectively, then we have the following uncertainty inequality*

$$\mathbf{H}^{\mathbf{X}\|\mathbf{Z}_1}(\mathbf{v}) + \mathbf{H}^{\mathbf{Y}\|\mathbf{Z}_2}(\mathbf{v}) \geq \max\left\{\log\left\{\left(|\varsigma_1|^2|\varsigma_2|^2\right)^n \left(\frac{1}{|\zeta|^2}\right)^{n\left(|\xi_1|^2+|\xi_2|^2\right)}\right\}, 0\right\}, \tag{9}$$

where the relative entropy $\mathbf{H}^{\mathbf{X}\|\mathbf{Z}_1}(\mathbf{v}) = \sum_i^n |\mathbf{x}_i^\dagger \mathbf{v}|^2 \log\frac{|\mathbf{x}_i^\dagger \mathbf{v}|^2}{|\mathbf{z}_{1,i}^\dagger \mathbf{v}|^2}$, $\mathbf{H}^{\mathbf{Y}\|\mathbf{Z}_2}(\mathbf{v}) = \sum_i^n |\mathbf{y}_i^\dagger \mathbf{v}|^2 \log\frac{|\mathbf{y}_i^\dagger \mathbf{v}|^2}{|\mathbf{z}_{2,i}^\dagger \mathbf{v}|^2}$, and $\zeta = \max_{i,j}\left\{|\mathbf{z}_{1,i}^\dagger \mathbf{v}|, |\mathbf{z}_{2,j}^\dagger \mathbf{v}|\right\}$, $\xi_1 = \max_i(|\mathbf{x}_i^\dagger \mathbf{v}|)$ and $\xi_2 = \max_i(|\mathbf{y}_i^\dagger \mathbf{v}|)$, $\varsigma_1 = \min_i(|\mathbf{x}_i^\dagger \mathbf{v}|) > 0$ and $\varsigma_2 = \min_i(|\mathbf{y}_i^\dagger \mathbf{v}|) > 0$.

Proof. from the definition of relative entropy, we know

$$\begin{aligned}\mathbf{H}^{\mathbf{X}\|\mathbf{Z}_1}(\mathbf{v}) &= \sum_i^n |\mathbf{x}_i^\dagger \mathbf{v}|^2 \log\frac{|\mathbf{x}_i^\dagger \mathbf{v}|^2}{|\mathbf{z}_{1,i}^\dagger \mathbf{v}|^2} \\ &= \sum_i^n |\mathbf{x}_i^\dagger \mathbf{v}|^2 \log|\mathbf{x}_i^\dagger \mathbf{v}|^2 - \sum_i^n |\mathbf{x}_i^\dagger \mathbf{v}|^2 \log|\mathbf{z}_{1,i}^\dagger \mathbf{v}|^2 \\ &= -\mathbf{H}^{\mathbf{X}}(\mathbf{v}) - \sum_i^n |\mathbf{x}_i^\dagger \mathbf{v}|^2 \log|\mathbf{z}_{1,i}^\dagger \mathbf{v}|^2.\end{aligned}$$

Thus we can obtain the following relations

$$\begin{aligned}\mathbf{H}^{\mathbf{X}\|\mathbf{Z}_1}(\mathbf{v}) + \mathbf{H}^{\mathbf{Y}\|\mathbf{Z}_2}(\mathbf{v}) &= -\mathbf{H}^{\mathbf{X}}(\mathbf{v}) - \sum_i^n |\mathbf{x}_i^\dagger \mathbf{v}|^2 \log|\mathbf{z}_{1,i}^\dagger \mathbf{v}|^2 - \mathbf{H}^{\mathbf{Y}}(\mathbf{v}) \\ &\quad - \sum_i^n |\mathbf{y}_i^\dagger \mathbf{v}|^2 \log|\mathbf{z}_{2,i}^\dagger \mathbf{v}|^2 \\ &= -\left[\mathbf{H}^{\mathbf{X}}(\mathbf{v}) + \mathbf{H}^{\mathbf{Y}}(\mathbf{v})\right] - \sum_i^n |\mathbf{x}_i^\dagger \mathbf{v}|^2 \log|\mathbf{z}_{1,i}^\dagger \mathbf{v}|^2 - \sum_i^n |\mathbf{y}_i^\dagger \mathbf{v}|^2 \log|\mathbf{z}_{2,i}^\dagger \mathbf{v}|^2.\end{aligned}$$

Taking into account the relation $\zeta = \max_{i,j}\left\{|\mathbf{z}_{1,i}^\dagger \mathbf{v}|, |\mathbf{z}_{2,j}^\dagger \mathbf{v}|\right\}$, then

$$\begin{aligned}\mathbf{H}^{\mathbf{X}\|\mathbf{Z}_1}(\mathbf{v}) + \mathbf{H}^{\mathbf{Y}\|\mathbf{Z}_2}(\mathbf{v}) &\geq -\left[\mathbf{H}^{\mathbf{X}}(\mathbf{v}) + \mathbf{H}^{\mathbf{Y}}(\mathbf{v})\right] - \sum_i^n \left(|\mathbf{x}_i^\dagger \mathbf{v}|^2 + |\mathbf{y}_i^\dagger \mathbf{v}|^2\right)\log|\zeta|^2 \\ &= -\left[\mathbf{H}^{\mathbf{X}}(\mathbf{v}) + \mathbf{H}^{\mathbf{Y}}(\mathbf{v})\right] - \log|\zeta|^2 \cdot \sum_i^n \left(|\mathbf{x}_i^\dagger \mathbf{v}|^2 + |\mathbf{y}_i^\dagger \mathbf{v}|^2\right).\end{aligned}$$

At the same time, we know $\xi_1 = \max_i(|\mathbf{x}_i^\dagger \mathbf{v}|)$ and $\xi_2 = \max_i(|\mathbf{y}_i^\dagger \mathbf{v}|)$, then the following inequality holds true

$$\begin{aligned}\mathbf{H}^{\mathbf{X}\|\mathbf{Z}_1}(\mathbf{v}) + \mathbf{H}^{\mathbf{Y}\|\mathbf{Z}_2}(\mathbf{v}) &\geq -\left[\mathbf{H}^{\mathbf{X}}(\mathbf{v}) + \mathbf{H}^{\mathbf{Y}}(\mathbf{v})\right] - \log|\zeta|^2 \cdot \sum_i^n \left(|\xi_1|^2 + |\xi_2|^2\right) \\ &\geq n\left(|\xi_1|^2 + |\xi_2|^2\right) \cdot \log\frac{1}{|\zeta|^2} - \left[\mathbf{H}^{\mathbf{X}}(\mathbf{v}) + \mathbf{H}^{\mathbf{Y}}(\mathbf{v})\right].\end{aligned}$$

On the other hand, considering the two parameters $\varsigma_1 = \min_i(|\mathbf{x}_i^\dagger \mathbf{v}|) > 0$ and $\varsigma_2 = \min_i(|\mathbf{y}_i^\dagger \mathbf{v}|) > 0$, we obtain the two relations as follows:

$$\mathbf{H}^{\mathbf{X}}(\mathbf{v}) = -\sum_i^n |\mathbf{x}_i^\dagger \mathbf{v}|^2 \log|\mathbf{x}_i^\dagger \mathbf{v}|^2 \leq -\sum_i^n |\mathbf{x}_i^\dagger \mathbf{v}|^2 \log|\varsigma_1|^2 = n \cdot \log\frac{1}{|\varsigma_1|^2},$$

$$\mathbf{H}^{\mathbf{Y}}(\mathbf{v}) = -\sum_i^n |\mathbf{y}_i^\dagger \mathbf{v}|^2 \log|\mathbf{y}_i^\dagger \mathbf{v}|^2 \leq -\sum_i^n |\mathbf{x}_i^\dagger \mathbf{v}|^2 \log|\varsigma_2|^2 = n \cdot \log\frac{1}{|\varsigma_2|^2}.$$

Therefore

$$\mathbf{H}^{\mathbf{X}\|\mathbf{Z}_1}(\mathbf{v}) + \mathbf{H}^{\mathbf{Y}\|\mathbf{Z}_2}(\mathbf{v}) \geq n\left(|\xi_1|^2 + |\xi_2|^2\right) \cdot \log\frac{1}{|\zeta|^2} - n\left(\log\frac{1}{|\varsigma_1|^2} + \log\frac{1}{|\varsigma_2|^2}\right)$$

Thus we have the following relation

$$\mathbf{H}^{\mathbf{X}\|\mathbf{Z}_1}(\mathbf{v}) + \mathbf{H}^{\mathbf{Y}\|\mathbf{Z}_2}(\mathbf{v}) \geq \log\left\{\left(|\varsigma_1|^2|\varsigma_2|^2\right)^n \left(\frac{1}{|\zeta|^2}\right)^{n\left(|\xi_1|^2+|\xi_2|^2\right)}\right\}.$$

Since the relative entropy is non-negative, the final result is obtained. The proof is finished.

From Theorem 1, this derived uncertainty relation shows the relation between the new bound and the parameters: the length n of the vector $\mathbf{v}$, the min non-zero correlation coefficients ς_1 and ς_2, and the max correlation coefficients ζ, ξ_1 and ξ_2.

As shown before in above section, the relative entropy could be used to measure the uncertainty at the same time with more priori knowledge (the given reference basis $\mathbf{Z}_1, \mathbf{Z}_2$). Therefore, in advance we could give the reference basis according to our wish. For example, if $\zeta = \max_{i,j}\left\{|\mathbf{z}_{1,i}^\dagger \mathbf{v}|, |\mathbf{z}_{2,j}^\dagger \mathbf{v}|\right\} = 1$ (that is to say, the vector signal is one single basis vector), then we have the following uncertainty relation of (10)

$$\mathbf{H}^{\mathbf{X}\|\mathbf{Z}_1}(\mathbf{v}) + \mathbf{H}^{\mathbf{Y}\|\mathbf{Z}_2}(\mathbf{v}) \geq \log\left(|\varsigma_1|^2|\varsigma_2|^2\right)^n, \tag{10}$$

which is only associated with ς_1 and ς_2. That is to say, at this time, the uncertainty bound is only related with the min non-zero correlation of the two series of basis (**X**,**Y**) with the vector **v**.

Specially, if $|\mathbf{z}_i^\dagger \mathbf{v}| \equiv |\mathbf{x}_i^\dagger \mathbf{v}|$, then we have

$$\mathbf{H}^{\mathbf{X}\|\mathbf{Z}_1}(\mathbf{v}) + \mathbf{H}^{\mathbf{Y}\|\mathbf{Z}_2}(\mathbf{v}) = \mathbf{H}^{\mathbf{Y}\|\mathbf{Z}}(\mathbf{v}) = \mathbf{H}^{\mathbf{Y}}(\mathbf{v}) - \sum_i^n |\mathbf{y}_i^\dagger \mathbf{v}|^2 \log|\mathbf{z}_{2,i}^\dagger \mathbf{v}|^2.$$

From the proof in Theorem 1, at this time we know that

$$\begin{aligned}\mathbf{H}^{\mathbf{X}\|\mathbf{Z}_1}(\mathbf{v}) + \mathbf{H}^{\mathbf{Y}\|\mathbf{Z}_2}(\mathbf{v}) &\geq -\left[\mathbf{H}^{\mathbf{X}}(\mathbf{v}) + \mathbf{H}^{\mathbf{Y}}(\mathbf{v})\right] - \log|\zeta|^2 \cdot \sum_i^n \left(|\xi_1|^2 + |\xi_2|^2\right) \\ &\geq n\left(|\xi_1|^2 + |\xi_2|^2\right) \cdot \log\frac{1}{|\zeta|^2} - \left[\mathbf{H}^{\mathbf{X}}(\mathbf{v}) + \mathbf{H}^{\mathbf{Y}}(\mathbf{v})\right]. \\ &\geq n\left(|\xi_1|^2 + |\xi_2|^2\right) \cdot \log\frac{1}{|\zeta|^2} - n\left(\log\frac{1}{|\varsigma_1|^2} + \log\frac{1}{|\varsigma_2|^2}\right) \\ &= n|\xi_1|^2 \cdot \log\frac{1}{|\zeta|^2} - n\log\frac{1}{|\varsigma_1|^2}\end{aligned}$$

$$= \log\left\{\left(|\varsigma_1|^2\right)^n\left(\frac{1}{|\zeta|^2}\right)^{n|\xi_1|^2}\right\},$$

i.e., $\mathbf{H}^{\mathbf{X}\|\mathbf{Z}_1}(\mathbf{v}) + \mathbf{H}^{\mathbf{Y}\|\mathbf{Z}_2}(\mathbf{v}) \geq \log\left\{\left(|\varsigma_1|^2\right)^n\left(\frac{1}{|\zeta|^2}\right)^{n|\xi_1|^2}\right\}$ while $|\mathbf{z}_i^\dagger\mathbf{v}| \equiv |\mathbf{x}_i^\dagger\mathbf{v}|$.

Obviously, $\log\left\{\left(|\varsigma_1|^2\right)^n\left(\frac{1}{|\zeta|^2}\right)^{n|\xi_1|^2}\right\} < \log\left\{\left(|\varsigma_1|^2|\varsigma_2|^2\right)^n\left(\frac{1}{|\zeta|^2}\right)^{n\left(|\xi_1|^2+|\xi_2|^2\right)}\right\}$ always holds true. That is to say, if the known (priori) basis ($\mathbf{Z}$) and one series of client basis ($\mathbf{X}$ or $\mathbf{Y}$) is the same (fully overlapped), then the uncertainty bound must be lower. In the same manner, if the known (priori) basis ($\mathbf{Z}$) and one series of client basis ($\mathbf{X}$ or $\mathbf{Y}$) is partly overlapped, the uncertainty bound would reduce as well according to the overlapping degree.

Specially, if we let $|\mathbf{z}_{1,i}^\dagger\mathbf{v}| = |\mathbf{z}_{2,i}^\dagger\mathbf{v}| \equiv \frac{1}{\sqrt{n}}$, then we have the following lemma.

Lemma 1. *For the given graph $\mathbf{G}_1$ with its vector signal $\mathbf{v} = [v_1\ v_2 \cdots v_n]^T$ satisfying $\mathrm{tr}\left((\mathbf{v}\mathbf{v}^\dagger)^2\right) = 1$, if we have three series of standard orthogonal graph basis $\mathbf{X} = [\mathbf{x}_1\ \mathbf{x}_2 \cdots \mathbf{x}_n]$,$\mathbf{Y} = [\mathbf{y}_1\ \mathbf{y}_2 \cdots \mathbf{y}_n]$,$\mathbf{Z}_1 = [\mathbf{z}_{1,1}\ \mathbf{z}_{1,2} \cdots \mathbf{z}_{1,n}]$, $\mathbf{Z}_2 = [\mathbf{z}_{2,1}\ \mathbf{z}_{2,2} \cdots \mathbf{z}_{2,n}]$ and two unitary transform matrix $\mathbf{U}_1, \mathbf{U}_2$ satisfying $\mathbf{X} = \mathbf{U}_1\mathbf{Y}, \mathbf{Y} = \mathbf{U}_2\mathbf{Z}$, $\mathbf{X}$ is the graph Fourier basis of the graph $\mathbf{G}_1$, and $\mathbf{Z}_1, \mathbf{Z}_2$is the reference Fourier basis of graph $\mathbf{G}_2, \mathbf{G}_3$, respectively, then we have the following uncertainty inequality*

$$\mathbf{H}^{\mathbf{X}\|\mathbf{Z}_1}(\mathbf{v}) + \mathbf{H}^{\mathbf{Y}\|\mathbf{Z}_2}(\mathbf{v}) \geq \max\left\{\log\left[n^2\left(|\varsigma_1|^2|\varsigma_2|^2\right)^n\right], 0\right\}, \tag{11}$$

where the relative entropy $\mathbf{H}^{\mathbf{X}\|\mathbf{Z}_1}(\mathbf{v}) = \sum_i^n|\mathbf{x}_i^\dagger\mathbf{v}|^2\log\frac{|\mathbf{x}_i^\dagger\mathbf{v}|^2}{|\mathbf{z}_{1,i}^\dagger\mathbf{v}|^2}$, $\mathbf{H}^{\mathbf{Y}\|\mathbf{Z}_2}(\mathbf{v}) = \sum_i^n|\mathbf{y}_i^\dagger\mathbf{v}|^2\log\frac{|\mathbf{y}_i^\dagger\mathbf{v}|^2}{|\mathbf{z}_{2,i}^\dagger\mathbf{v}|^2}$, and $|\mathbf{z}_{1,i}^\dagger\mathbf{v}| = |\mathbf{z}_{2,i}^\dagger\mathbf{v}| \equiv \frac{1}{\sqrt{n}}$, $\xi_1 = \max_i\left(|\mathbf{x}_i^\dagger\mathbf{v}|\right)$ and $\xi_2 = \max_i\left(|\mathbf{y}_i^\dagger\mathbf{v}|\right)$, $\varsigma_1 = \min_i\left(|\mathbf{x}_i^\dagger\mathbf{v}|\right)$ and $\varsigma_2 = \min_i\left(|\mathbf{y}_i^\dagger\mathbf{v}|\right)$.

Proof. from the proof of Theorem 1, we have

$$\begin{aligned}\mathbf{H}^{\mathbf{X}\|\mathbf{Z}_1}(\mathbf{v}) + \mathbf{H}^{\mathbf{Y}\|\mathbf{Z}_2}(\mathbf{v}) &= -\mathbf{H}^{\mathbf{X}}(\mathbf{v}) - \sum_i^n|\mathbf{x}_i^\dagger\mathbf{v}|^2\log\left|\mathbf{z}_{1,i}^\dagger\mathbf{v}\right|^2 - \mathbf{H}^{\mathbf{Y}}(\mathbf{v}) \\ &\quad - \sum_i^n|\mathbf{y}_i^\dagger\mathbf{v}|^2\log\left|\mathbf{z}_{2,i}^\dagger\mathbf{v}\right|^2 \\ &= -\mathbf{H}^{\mathbf{X}}(\mathbf{v}) - \mathbf{H}^{\mathbf{Y}}(\mathbf{v}) - 2\log\frac{1}{n} \\ &\geq -n\left(\log\frac{1}{|\varsigma_1|^2} + \log\frac{1}{|\varsigma_2|^2}\right) + \log n^2 \\ &= \log\left[n^2\left(|\varsigma_1|^2|\varsigma_2|^2\right)^n\right].\end{aligned}$$

The proof is completed and trivial.

That is to say, if we let $|\mathbf{z}_{1,i}^\dagger\mathbf{v}| = |\mathbf{z}_{2,i}^\dagger\mathbf{v}| \equiv \frac{1}{\sqrt{n}}$, then this derived uncertainty relation shows the relation between the new bound and the parameters: the length n of the vector $\mathbf{v}$, the min non-zero correlation coefficients ς_1 and ς_2.

Generally, $\mathbf{Z}_1 \neq \mathbf{Z}_2$, at this time we find that it is possible that $\mathbf{H}^{\mathbf{X}\|\mathbf{Z}_1}(\mathbf{v}) + \mathbf{H}^{\mathbf{Y}\|\mathbf{Z}_2}(\mathbf{v}) = 0$, as will be our the most ideal case for zero bound. Instead, if $\mathbf{Z}_1 = \mathbf{Z}_2$, that is to say, the two given reference basis are the same (or the two given reference distributions are the same), then the uncertainty bound cannot reduce to zero. Or to say, after giving a reference basis including corresponding signal energy distribution, then any signal in two transform domains cannot have the same distribution any more as that of the given basis, which is the new extension of traditional entropic uncertainty relations without relative entropy.

4. Relative entropy based uncertainty principles on graph matrix

Now in this section we will discuss the case of graph matrix signal.

Theorem 2. *Assume that we have three graphs ($\mathbf{G}_1$, $\mathbf{G}_2, \mathbf{G}_3$) and client graph $\mathbf{G}_1$ has some similar graph structure and graph signal with that of reference graph $\mathbf{G}_2, \mathbf{G}_3$. For the given client graph $\mathbf{G}_1$ with its matrix signal $\mathbf{V} = \sum_k^K\lambda_k\mathbf{v}_k\mathbf{v}_k^\dagger$ ($\lambda_k > 0$) with $\mathrm{tr}\left(\mathbf{V}^2\right) < 1$ and $\mathrm{tr}(\mathbf{V}) = 1$, if we have four series of standard orthogonal basis $\mathbf{X} = [\mathbf{x}_1\ \mathbf{x}_2 \ldots \mathbf{x}_n]$,$\mathbf{Y} = [\mathbf{y}_1\ \mathbf{y}_2 \ldots \mathbf{y}_n]$ and $\mathbf{Z}_1 = [\mathbf{z}_{1,1}\ \mathbf{z}_{1,2} \cdots \mathbf{z}_{1,n}]$, $\mathbf{Z}_2 = [\mathbf{z}_{2,1}\ \mathbf{z}_{2,2} \cdots \mathbf{z}_{2,n}]$, and two unitary transform matrix $\mathbf{U}_1, \mathbf{U}_2$ with $\mathbf{Y} = \mathbf{U}_1\mathbf{X}$ and $\mathbf{X} = \mathbf{U}_2\mathbf{Z}$, $\mathbf{X}$ is the graph Fourier basis of the graph $\mathbf{G}_1$, and $\mathbf{Z}_1, \mathbf{Z}_2$is the reference Fourier basis of graph $\mathbf{G}_2, \mathbf{G}_3$, respectively, then we have the following uncertainty relation*

$$\mathbf{H}^{\mathbf{X}\|\mathbf{Z}_1}(\mathbf{V}) + \mathbf{H}^{\mathbf{Y}\|\mathbf{Z}_2}(\mathbf{V}) \geq \max\left\{0, \log\left\{\left(|\varsigma_1|^2|\varsigma_2|^2\right)^{n\mathrm{K}\lambda_{\min}}\left(\frac{1}{|\zeta|^2}\right)^{n\mathrm{K}\lambda_{\min}\left(|\xi_1|^2+|\xi_2|^2\right)}\right\}\right\}, \tag{12}$$

where the relative entropy $\mathbf{H}^{\mathbf{X}\|\mathbf{Z}_1}(\mathbf{V}) = \sum_{k=1}^{\mathrm{K}}\left(\lambda_k\sum_{i-1}^n|\mathbf{x}_i^\dagger\mathbf{v}_k|^2\log\frac{|\mathbf{x}_i^\dagger\mathbf{v}_k|^2}{|\mathbf{z}_{1,i}^\dagger\mathbf{v}_k|^2}\right)$, $\mathbf{H}^{\mathbf{Y}\|\mathbf{Z}_2}(\mathbf{V}) = \sum_{k=1}^{\mathrm{K}}\left(\lambda_k\sum_{i-1}^n|\mathbf{y}_i^\dagger\mathbf{v}_k|^2\log\frac{|\mathbf{y}_i^\dagger\mathbf{v}_k|^2}{|\mathbf{z}_{2,i}^\dagger\mathbf{v}_k|^2}\right)$, and $\zeta = \max_{i,j,k}\left\{|\mathbf{z}_{1,i}^\dagger\mathbf{v}_k|, |\mathbf{z}_{2,j}^\dagger\mathbf{v}_k|\right\}$, $\xi_1 = \max_{i,k}\left(|\mathbf{x}_i^\dagger\mathbf{v}_k|\right)$ and $\xi_2 = \max_{i,k}\left(|\mathbf{y}_i^\dagger\mathbf{v}_k|\right)$, $\lambda_{\min} = \min_k\{\lambda_k \neq 0\}$, $\varsigma_1 = \min_{i,k}\left(|\mathbf{x}_i^\dagger\mathbf{v}_k|\right) > 0$ and $\varsigma_2 = \min_{i,k}\left(|\mathbf{y}_i^\dagger\mathbf{v}_k|\right) > 0$.

Proof. from the definition of relative entropy of matrix, we have

$$\mathbf{H}^{\mathbf{X}\|\mathbf{Z}_1}(\mathbf{V}) = \sum_{k=1}^{\mathrm{K}}\left(\lambda_k\sum_{i-1}^n|\mathbf{x}_i^\dagger\mathbf{v}_k|^2\log\frac{|\mathbf{x}_i^\dagger\mathbf{v}_k|^2}{|\mathbf{z}_{1,i}^\dagger\mathbf{v}_k|^2}\right) = \sum_{k=1}^{\mathrm{K}}\left(\lambda_k\mathbf{H}^{\mathbf{X}\|\mathbf{Z}_1}(\mathbf{v}_k)\right),$$

And in the same manner, we have the following relation

$$\mathbf{H}^{\mathbf{Y}\|\mathbf{Z}_2}(\mathbf{V}) = \sum_{k=1}^{\mathrm{K}}\left(\lambda_k\mathbf{H}^{\mathbf{Y}\|\mathbf{Z}_2}(\mathbf{v}_k)\right).$$

Therefore, we obtain the following relation

$$\begin{aligned}\mathbf{H}^{\mathbf{X}\|\mathbf{Z}_1}(\mathbf{V}) + \mathbf{H}^{\mathbf{Y}\|\mathbf{Z}_2}(\mathbf{V}) &= \sum_{k=1}^{\mathrm{K}}\left(\lambda_k\mathbf{H}^{\mathbf{X}\|\mathbf{Z}_1}(\mathbf{v}_k)\right) + \sum_{k=1}^{\mathrm{K}}\left(\lambda_k\mathbf{H}^{\mathbf{Y}\|\mathbf{Z}_2}(\mathbf{v}_k)\right) \\ &= \sum_{k=1}^{\mathrm{K}}\lambda_k\left(\mathbf{H}^{\mathbf{X}\|\mathbf{Z}_1}(\mathbf{v}_k) + \mathbf{H}^{\mathbf{Y}\|\mathbf{Z}_2}(\mathbf{v}_k)\right).\end{aligned}$$

From Theorem 1, we know the two relations hold true

$$\mathbf{H}^{\mathbf{X}\|\mathbf{Z}_1}(\mathbf{v}_k) = -\mathbf{H}^{\mathbf{X}}(\mathbf{v}_k) - \sum_i^n|\mathbf{x}_i^\dagger\mathbf{v}_k|^2\log\left|\mathbf{z}_{1,i}^\dagger\mathbf{v}_k\right|^2,$$

$$\mathbf{H}^{\mathbf{Y}\|\mathbf{Z}_2}(\mathbf{v}_k) = -\mathbf{H}^{\mathbf{Y}}(\mathbf{v}_k) - \sum_i^n |\mathbf{y}_i^\dagger \mathbf{v}_k|^2 \log\left|\mathbf{z}_{2,i}^\dagger \mathbf{v}_k\right|^2.$$

Therefore, taking into account of the relation $\zeta = \max_{i,j,k}\left\{|\mathbf{z}_{1,i}^\dagger \mathbf{v}_k|, |\mathbf{z}_{2,j}^\dagger \mathbf{v}_k|\right\}$, $\xi_1 = \max_{i,k}(|\mathbf{x}_i^\dagger \mathbf{v}_k|)$ and $\xi_2 = \max_{i,k}(|\mathbf{y}_i^\dagger \mathbf{v}_k|)$, we get the inequality

$$\begin{aligned}
&\mathbf{H}^{\mathbf{X}\|\mathbf{Z}_1}(\mathbf{v}_k) + \mathbf{H}^{\mathbf{Y}\|\mathbf{Z}_2}(\mathbf{v}_k) \\
&\geq -\left[\mathbf{H}^{\mathbf{X}}(\mathbf{v}_k) + \mathbf{H}^{\mathbf{Y}}(\mathbf{v}_k)\right] - \sum_i^n \left(|\mathbf{x}_i^\dagger \mathbf{v}_k|^2 + |\mathbf{y}_i^\dagger \mathbf{v}_k|^2\right)\log|\zeta|^2 \\
&\geq -\left[\mathbf{H}^{\mathbf{X}}(\mathbf{v}) + \mathbf{H}^{\mathbf{Y}}(\mathbf{v})\right] - \log|\zeta|^2 \cdot \sum_i^n \left(|\xi_1|^2 + |\xi_2|^2\right) \\
&\geq n\left(|\xi_1|^2 + |\xi_2|^2\right)\cdot \log\frac{1}{|\zeta|^2} - \left[\mathbf{H}^{\mathbf{X}}(\mathbf{v}) + \mathbf{H}^{\mathbf{Y}}(\mathbf{v})\right].
\end{aligned}$$

On the other hand, from $\varsigma_1 = \min_{i,k}(|\mathbf{x}_i^\dagger \mathbf{v}_k|) > 0$ and $\varsigma_2 = \min_{i,k}(|\mathbf{y}_i^\dagger \mathbf{v}_k|) > 0$

$$\mathbf{H}^{\mathbf{X}}(\mathbf{v}_k) = -\sum_i^n |\mathbf{x}_i^\dagger \mathbf{v}_k|^2 \log|\mathbf{x}_i^\dagger \mathbf{v}_k|^2 \leq -\sum_i^n |\mathbf{x}_i^\dagger \mathbf{v}_k|^2 \log|\varsigma_1|^2 = n\cdot\log\frac{1}{|\varsigma_1|^2},$$

$$\mathbf{H}^{\mathbf{Y}}(\mathbf{v}_k) = -\sum_i^n |\mathbf{y}_i^\dagger \mathbf{v}_k|^2 \log|\mathbf{y}_i^\dagger \mathbf{v}_k|^2 \leq -\sum_i^n |\mathbf{x}_i^\dagger \mathbf{v}_k|^2 \log|\varsigma_2|^2 = n\cdot\log\frac{1}{|\varsigma_2|^2}.$$

So the following relation holds true

$$\begin{aligned}
\mathbf{H}^{\mathbf{X}\|\mathbf{Z}_1}(\mathbf{v}_k) + \mathbf{H}^{\mathbf{Y}\|\mathbf{Z}_2}(\mathbf{v}_k) \geq\ & n\left(|\xi_1|^2 + |\xi_2|^2\right)\cdot\log\frac{1}{|\zeta|^2} \\
& - n\left(\log\frac{1}{|\varsigma_1|^2} + \log\frac{1}{|\varsigma_2|^2}\right).
\end{aligned}$$

Thus we obtain the inequality as follows

$$\begin{aligned}
&\mathbf{H}^{\mathbf{X}\|\mathbf{Z}_1}(\mathbf{V}) + \mathbf{H}^{\mathbf{Y}\|\mathbf{Z}_2}(\mathbf{V}) \\
&= \sum_{k=1}^{\mathrm{K}} \lambda_k \left(\mathbf{H}^{\mathbf{X}\|\mathbf{Z}_1}(\mathbf{v}_k) + \mathbf{H}^{\mathbf{Y}\|\mathbf{Z}_2}(\mathbf{v}_k)\right) \\
&\geq \sum_{k=1}^{\mathrm{K}} \lambda_k \left(n\left(|\xi_1|^2 + |\xi_2|^2\right)\cdot\log\frac{1}{|\zeta|^2} - n\left(\log\frac{1}{|\varsigma_1|^2} + \log\frac{1}{|\varsigma_2|^2}\right)\right).
\end{aligned}$$

Because $\lambda_{\min} = \min_k\{\lambda_k \neq 0\}$, we can get the final result as follows

$$\mathbf{H}^{\mathbf{X}\|\mathbf{Z}_1}(\mathbf{V}) + \mathbf{H}^{\mathbf{Y}\|\mathbf{Z}_2}(\mathbf{V}) \geq K\lambda_{\min}\left(n\left(|\xi_1|^2 + |\xi_2|^2\right)\cdot\log\frac{1}{|\zeta|^2} - n\left(\log\frac{1}{|\varsigma_1|^2} + \log\frac{1}{|\varsigma_2|^2}\right)\right)$$

$$\geq \log\left\{\left(|\varsigma_1|^2|\varsigma_2|^2\right)^{nK\lambda_{\min}}\left(\frac{1}{|\zeta|^2}\right)^{nK\lambda_{\min}\left(|\xi_1|^2+|\xi_2|^2\right)}\right\}.$$

The proof is completed.

From Theorem 2, this derived uncertainty relation shows the relation between the new bound and the parameters: the length n of the vector $\mathbf{v}$, the number of eigenvalues of matrix: K, the min non-zero correlation coefficients ς_1 and ς_2, and the max correlation coefficients ζ, ξ_1 and ξ_2, the min non-zero eigenvalue $\lambda_{\min}$.

Specially, if $K = 1$, then $\mathbf{H}^{\mathbf{X}\|\mathbf{Z}_1}(\mathbf{V}) = \mathbf{H}^{\mathbf{X}\|\mathbf{Z}_1}(\mathbf{v})$ and $\mathbf{H}^{\mathbf{Y}\|\mathbf{Z}_2}(\mathbf{V}) = \mathbf{H}^{\mathbf{Y}\|\mathbf{Z}_2}(\mathbf{v})$, which reduces to the case of vector. On the other hand, if $L = \|\lambda_k\|_0 < K$, then we have the following lemma.

Lemma 2. *For a given graph $\mathbf{G}_1$ with its matrix signal $\mathbf{V} = \sum_k^K \lambda_k \mathbf{v}_k \mathbf{v}_k^\dagger$ ($\lambda_k > 0$) with $\mathrm{tr}(\mathbf{V}^2) < 1$ and $\mathrm{tr}(\mathbf{V}) = 1$, if we have three series of standard orthogonal basis $\mathbf{X} = [\mathbf{x}_1\ \mathbf{x}_2 \ldots \mathbf{x}_n]$, $\mathbf{Y} = [\mathbf{y}_1\ \mathbf{y}_2 \ldots \mathbf{y}_n]$ and $\mathbf{Z}_1 = [\mathbf{z}_{1,1}\ \mathbf{z}_{1,2} \cdots \mathbf{z}_{1,n}]$, $\mathbf{Z}_2 = [\mathbf{z}_{2,1}\ \mathbf{z}_{2,2} \cdots \mathbf{z}_{2,n}]$, and two unitary transform matrix $\mathbf{U}_1, \mathbf{U}_2$ with $\mathbf{Y} = \mathbf{U}_1\mathbf{X}$ and $\mathbf{X} = \mathbf{U}_2\mathbf{Z}$, $\mathbf{X}$ is the graph Fourier basis of the graph $\mathbf{G}_1$, and $\mathbf{Z}_1, \mathbf{Z}_2$ is the reference Fourier basis of graph $\mathbf{G}_2, \mathbf{G}_3$, respectively, then we have the following uncertainty relation*

$$\begin{aligned}
&\mathbf{H}^{\mathbf{X}\|\mathbf{Z}_1}(\mathbf{V}) + \mathbf{H}^{\mathbf{Y}\|\mathbf{Z}_2}(\mathbf{V}) \\
&\qquad \geq \max\left\{0,\ \log\left\{\left(|\varsigma_1|^2|\varsigma_2|^2\right)^{nL\lambda_{\min}}\left(\frac{1}{|\zeta|^2}\right)^{nL\lambda_{\min}\left(|\xi_1|^2+|\xi_2|^2\right)}\right\}\right\},
\end{aligned} \tag{13}$$

where $L = \|\lambda_k\|_0 < K$, the relative entropy $\mathbf{H}^{\mathbf{X}\|\mathbf{Z}_1}(\mathbf{V}) = \sum_{k=1}^{\mathrm{K}}\left(\lambda_k \sum_{i-1}^n |\mathbf{x}_i^\dagger \mathbf{v}_k|^2 \log\frac{|\mathbf{x}_i^\dagger \mathbf{v}_k|^2}{|\mathbf{z}_{1,i}^\dagger \mathbf{v}_k|^2}\right)$, $\mathbf{H}^{\mathbf{Y}\|\mathbf{Z}_2}(\mathbf{V}) = \sum_{k=1}^{\mathrm{K}}\left(\lambda_k \sum_{i-1}^n |\mathbf{y}_i^\dagger \mathbf{v}_k|^2 \log\frac{|\mathbf{y}_i^\dagger \mathbf{v}_k|^2}{|\mathbf{z}_{2,i}^\dagger \mathbf{v}_k|^2}\right)$, and $\zeta = \max_{i,j,k}\left\{|\mathbf{z}_{1,i}^\dagger \mathbf{v}_k|,\ |\mathbf{z}_{2,j}^\dagger \mathbf{v}_k|\right\}$, $\xi_1 = \max_{i,k}(|\mathbf{x}_i^\dagger \mathbf{v}_k|)$ and $\xi_2 = \max_{i,k}(|\mathbf{y}_i^\dagger \mathbf{v}_k|)$, $\lambda_{\min} = \min_k\{\lambda_k \neq 0\}$, $\varsigma_1 = \min_{i,k}(|\mathbf{x}_i^\dagger \mathbf{v}_k|) > 0$ and $\varsigma_2 = \min_{i,k}(|\mathbf{y}_i^\dagger \mathbf{v}_k|) > 0$.

Proof. from the proof of Theorem 2, we know

$$\begin{aligned}
&\mathbf{H}^{\mathbf{X}\|\mathbf{Z}_1}(\mathbf{v}_k) + \mathbf{H}^{\mathbf{Y}\|\mathbf{Z}_2}(\mathbf{v}_k) \\
&\quad \geq -\left[\mathbf{H}^{\mathbf{X}}(\mathbf{v}_k) + \mathbf{H}^{\mathbf{Y}}(\mathbf{v}_k)\right] - \sum_i^n \left(|\mathbf{x}_i^\dagger \mathbf{v}_k|^2 + |\mathbf{y}_i^\dagger \mathbf{v}_k|^2\right)\log|\zeta|^2 \quad \text{wiht} \quad \zeta = \max_{i,j,k}\left\{|\mathbf{z}_{1,i}^\dagger \mathbf{v}_k|, |\mathbf{z}_{2,j}^\dagger \mathbf{v}_k|\right\} \\
&\quad \geq -\left[\mathbf{H}^{\mathbf{X}}(\mathbf{v}) + \mathbf{H}^{\mathbf{Y}}(\mathbf{v})\right] - \log|\zeta|^2 \cdot \sum_i^n \left(|\xi_1|^2 + |\xi_2|^2\right) \quad \text{with} \quad \xi_1 = \max_{i,k}(|\mathbf{x}_i^\dagger \mathbf{v}_k|) \text{ and } \xi_2 = \max_{i,k}(|\mathbf{y}_i^\dagger \mathbf{v}_k|) \\
&\geq n\left(|\xi_1|^2 + |\xi_2|^2\right)\cdot\log\frac{1}{|\zeta|^2} - \left[\mathbf{H}^{\mathbf{X}}(\mathbf{v}) + \mathbf{H}^{\mathbf{Y}}(\mathbf{v})\right].
\end{aligned}$$

On the other hand, we know

$$\mathbf{H}^{\mathbf{X}\|\mathbf{Z}_1}(\mathbf{v}_k) + \mathbf{H}^{\mathbf{Y}\|\mathbf{Z}_2}(\mathbf{v}_k) \geq n\left(|\xi_1|^2 + |\xi_2|^2\right)\cdot\log\frac{1}{|\zeta|^2} - n\left(\log\frac{1}{|\varsigma_1|^2} + \log\frac{1}{|\varsigma_2|^2}\right).$$

Thus we obtain

$$\begin{aligned}
&\mathbf{H}^{\mathbf{X}\|\mathbf{Z}_1}(\mathbf{V}) + \mathbf{H}^{\mathbf{Y}\|\mathbf{Z}_2}(\mathbf{V}) \\
&= \sum_{k=1}^{\mathrm{K}} \lambda_k \left(\mathbf{H}^{\mathbf{X}\|\mathbf{Z}_1}(\mathbf{v}_k) + \mathbf{H}^{\mathbf{Y}\|\mathbf{Z}_2}(\mathbf{v}_k)\right) \\
&\geq \sum_{k=1}^{\mathrm{K}} \lambda_k \left(n\left(|\xi_1|^2 + |\xi_2|^2\right)\cdot\log\frac{1}{|\zeta|^2} - n\left(\log\frac{1}{|\varsigma_1|^2} + \log\frac{1}{|\varsigma_2|^2}\right)\right)
\end{aligned}$$

$$\geq L\lambda_{\min}\left(n\left(|\xi_1|^2+|\xi_2|^2\right)\cdot\log\frac{1}{|\zeta|^2}-n\left(\log\frac{1}{|\varsigma_1|^2}+\log\frac{1}{|\varsigma_2|^2}\right)\right) \quad \text{with} \quad L = \|\lambda_k\|_0 < K \text{ and } \lambda_{\min} = \min_k\{\lambda_k \neq 0\}.$$

$$\geq \log\left\{\left(|\varsigma_1|^2|\varsigma_2|^2\right)^{nL\lambda_{\min}}\left(\frac{1}{|\zeta|^2}\right)^{nL\lambda_{\min}\left(|\xi_1|^2+|\xi_2|^2\right)}\right\}.$$

The proof is completed.

From Lemma 2, this derived uncertainty relation shows the relation between the new bound and the parameters: the length n of the vector $\mathbf{v}$, the number of non-zero eigenvalues of matrix: L, the min non-zero correlation coefficients ς_1 and ς_2, and the max correlation coefficients ζ, ξ_1 and ξ_2, the min non-zero eigenvalue $\lambda_{\min}$. That is to say, if the matrix is sparse, the bound will reduce to lower value. Especially, if $L = 1$, then we have the following relation:

$$\mathbf{H}^{\mathbf{X}\|\mathbf{Z}_1}(\mathbf{V})+\mathbf{H}^{\mathbf{Y}\|\mathbf{Z}_2}(\mathbf{V}) \geq \log\left\{\left(|\varsigma_1|^2|\varsigma_2|^2\right)^{n\lambda_{\min}}\left(\frac{1}{|\zeta|^2}\right)^{n\lambda_{\min}\left(|\xi_1|^2+|\xi_2|^2\right)}\right\}.$$

In addition, if $\zeta = \max_{i,j,k}\left\{|\mathbf{z}_{1,i}^{\dagger}\mathbf{v}_k|, |\mathbf{z}_{2,j}^{\dagger}\mathbf{v}_k|\right\}$, then we will have

$$\mathbf{H}^{\mathbf{X}\|\mathbf{Z}_1}(\mathbf{V})+\mathbf{H}^{\mathbf{Y}\|\mathbf{Z}_2}(\mathbf{V}) \geq \log\left\{\left(|\varsigma_1|^2|\varsigma_2|^2\right)^{n\lambda_{\min}}\right\},$$

which is only related to the length n of vectors, the min non-zero correlation coefficients ς_1 and ς_2, and the min non-zero eigenvalue $\lambda_{\min}$.

Similarly, generally, $\mathbf{Z}_1 \neq \mathbf{Z}_2$, at this time we find that it is possible that $\mathbf{H}^{\mathbf{X}\|\mathbf{Z}_1}(\mathbf{V})+\mathbf{H}^{\mathbf{Y}\|\mathbf{Z}_2}(\mathbf{V}) = 0$, as will be our the most ideal case for zero bound in terms of matrix signal. Instead, if $\mathbf{Z}_1 = \mathbf{Z}_2$, that is to say, the two given reference basis are the same (or the two given reference distributions are the same) in terms of matrix signal, then the uncertainty bound cannot reduce to zero. Or to say, after giving a reference basis including corresponding signal energy distribution in terms of matrix signal, then any matrix signal in two transform domains cannot have the same distribution any more as that of the given basis, which is the new extension of traditional entropic uncertainty relations without relative entropy in terms of matrix signal.

5. Combined uncertainty principles

We must not that one fact is that relative entropy is employed to measure the distance between two signal energy distributions so that the value of relative entropy can be zero or $+\infty$. If the two signal energy distributions are the same, the value of relative entropy is zero. Instead, if the two signal energy distributions are fully different, then the value of relative entropy tend to $+\infty$, which is determined by the property of relative entropy. In other words, if the reference basis or the given reference signal energy distribution is fully different from that of the client signal, we will obtain the infinitely great quantity, which violates our original intention. We hope our uncertainty more and more less. Therefore, here we will combine the Shannon entropy to measure the uncertainty together. First see the following lemma.

Lemma 3. *For the given vector signal* $\mathbf{v} = [v_1\ v_2 \cdots v_n]^T$ *satisfying* $\mathrm{tr}(\mathbf{v}\mathbf{v}^{\dagger}) = 1$, *if we have two series of standard orthogonal basis* $\mathbf{X} = [\mathbf{x}_1\ \mathbf{x}_2 \cdots \mathbf{x}_n]$ *and* $\mathbf{Y} = [\mathbf{y}_1\ \mathbf{y}_2 \cdots \mathbf{y}_n]$, *then we have the following uncertainty inequality*

$$\mathbf{H}^{\mathbf{X}}(\mathbf{v})+\mathbf{H}^{\mathbf{Y}}(\mathbf{v}) \geq \ln\frac{1}{\xi_1\cdot\xi_2}, \tag{14}$$

where the Shannon entropy $\mathbf{H}^{\mathbf{X}}(\mathbf{v}) = \sum_i^n|\mathbf{x}_i^{\dagger}\mathbf{v}|^2\log\frac{1}{|\mathbf{x}_i^{\dagger}\mathbf{v}|^2}$, $\mathbf{H}^{\mathbf{Y}}(\mathbf{v}) = \sum_i^n|\mathbf{y}_i^{\dagger}\mathbf{v}|^2\log\frac{1}{|\mathbf{y}_i^{\dagger}\mathbf{v}|^2}$, and $\xi_{\max} = \max_{i,j}\left(|\mathbf{x}_i^{\dagger}\mathbf{y}_j|\right)$, $\xi_1 = \max_i\left(|\mathbf{x}_i^{\dagger}\mathbf{v}|\right)$ and $\xi_2 = \max_i\left(|\mathbf{y}_i^{\dagger}\mathbf{v}|\right)$.

Proof. From Theorem 1 in [52], the following inequality is easily obtained

$$\mathbf{H}^{\mathbf{X}}(\mathbf{v})+\mathbf{H}^{\mathbf{Y}}(\mathbf{v}) \geq \ln\frac{1}{\xi_{\max}} \text{ with } \xi_{\max} = \max_{i,j}\left(|\mathbf{x}_i^{\dagger}\mathbf{y}_j|\right).$$

At the same time, we know

$$\xi_{\max} = \max_{i,j}\left(\left|\mathbf{x}_i^{\dagger}\mathbf{y}_j\right|\right) = \max_{i,j}\left(\left|\mathbf{x}_i^{\dagger}\mathbf{v}\mathbf{v}^{\dagger}\mathbf{y}_j\right|\right) = \max_{i,j}\left(\left|\mathbf{x}_i^{\dagger}\mathbf{v}\mathbf{v}^{\dagger}\mathbf{y}_j\right|\right).$$

Therefore, we have

$$\xi_{\max} = \max_{i,j}\left\{\left|\left(\mathbf{x}_i^{\dagger}\mathbf{v}\right)\left(\mathbf{v}^{\dagger}\mathbf{y}_j\right)\right|\right\} = \max_j\left\{\left|\left(\mathbf{x}_i^{\dagger}\mathbf{v}\right)\right|\right\}\max_j\left\{\left|\left(\mathbf{v}^{\dagger}\mathbf{y}_j\right)\right|\right\}$$
$$= \max_j\left\{\left|\left(\mathbf{x}_i^{\dagger}\mathbf{v}\right)\right|\right\}\max_j\left\{\left|\left(\mathbf{y}_j^{\dagger}\mathbf{v}\right)\right|\right\}.$$

Taking into account of $\xi_1 = \max_i\left(|\mathbf{x}_i^{\dagger}\mathbf{v}|\right)$ and $\xi_2 = \max_i\left(|\mathbf{y}_i^{\dagger}\mathbf{v}|\right)$, thus we obtain the relation as follows

$$\xi_{\max} = \xi_1\cdot\xi_2.$$

Therefore, the lemma proof is completed.

Now we will combine this lemma with Theorem 1 to obtain our new combined uncertainty relation as follows:

Lemma 4. *For the given graph* $\mathbf{G}_1$ *with its vector signal* $\mathbf{v} = [v_1\ v_2 \cdots v_n]^T$ *satisfying* $\mathrm{tr}\left((\mathbf{v}\mathbf{v}^{\dagger})^2\right) = 1$, *if we have three series of standard orthogonal graph basis* $\mathbf{X} = [\mathbf{x}_1\ \mathbf{x}_2 \cdots \mathbf{x}_n]$, $\mathbf{Y} = [\mathbf{y}_1\ \mathbf{y}_2 \cdots \mathbf{y}_n]$, $\mathbf{Z}_1 = [\mathbf{z}_{1,1}\ \mathbf{z}_{1,2} \cdots \mathbf{z}_{1,n}]$, $\mathbf{Z}_2 = [\mathbf{z}_{2,1}\ \mathbf{z}_{2,2} \cdots \mathbf{z}_{2,n}]$ *and two unitary transform matrix* $\mathbf{U}_1, \mathbf{U}_2$ *satisfying* $\mathbf{X} = \mathbf{U}_1\mathbf{Y}$, $\mathbf{Y} = \mathbf{U}_2\mathbf{Z}$, $\mathbf{X}$ *is the graph Fourier basis of the graph* $\mathbf{G}_1$, *and* $\mathbf{Z}_1, \mathbf{Z}_2$ *is the reference Fourier basis of graph* $\mathbf{G}_2, \mathbf{G}_3$, *respectively, then we have the following uncertainty inequality*

$$\min\left\{\mathbf{H}^{\mathbf{X}\|\mathbf{Z}_1}(\mathbf{v}), \mathbf{H}^{\mathbf{X}}(\mathbf{v})\right\} + \min\left\{\mathbf{H}^{\mathbf{Y}\|\mathbf{Z}_2}(\mathbf{v}), \mathbf{H}^{\mathbf{Y}}(\mathbf{v})\right\}$$
$$\geq \min\left\{\ln\frac{1}{\xi_1\cdot\xi_2}, \max\left\{\log\left\{\left(|\varsigma_1|^2|\varsigma_2|^2\right)^n\left(\frac{1}{|\zeta|^2}\right)^{n\left(|\xi_1|^2+|\xi_2|^2\right)}\right\}, 0\right\}\right\}, \tag{15}$$

where the relative entropy $\mathbf{H}^{\mathbf{X}\|\mathbf{Z}_1}(\mathbf{v}) = \sum_i^n|\mathbf{x}_i^{\dagger}\mathbf{v}|^2\log\frac{|\mathbf{x}_i^{\dagger}\mathbf{v}|^2}{|\mathbf{z}_{1,i}^{\dagger}\mathbf{v}|^2}$, $\mathbf{H}^{\mathbf{Y}\|\mathbf{Z}_2}(\mathbf{v}) = \sum_i^n|\mathbf{y}_i^{\dagger}\mathbf{v}|^2\log\frac{|\mathbf{y}_i^{\dagger}\mathbf{v}|^2}{|\mathbf{z}_{2,i}^{\dagger}\mathbf{v}|^2}$, and $\zeta = \max_{i,j}\left\{|\mathbf{z}_{1,i}^{\dagger}\mathbf{v}|, |\mathbf{z}_{2,j}^{\dagger}\mathbf{v}|\right\}$, $\xi_{\max} = \max_{i,j}\left(|\mathbf{x}_i^{\dagger}\mathbf{y}_j|\right)$, $\xi_1 = \max_i\left(|\mathbf{x}_i^{\dagger}\mathbf{v}|\right)$ and $\xi_2 = \max_i\left(|\mathbf{y}_i^{\dagger}\mathbf{v}|\right)$, $\varsigma_1 = \min_i\left(|\mathbf{x}_i^{\dagger}\mathbf{v}|\right) > 0$ and $\varsigma_2 = \min_i\left(|\mathbf{y}_i^{\dagger}\mathbf{v}|\right) > 0$.

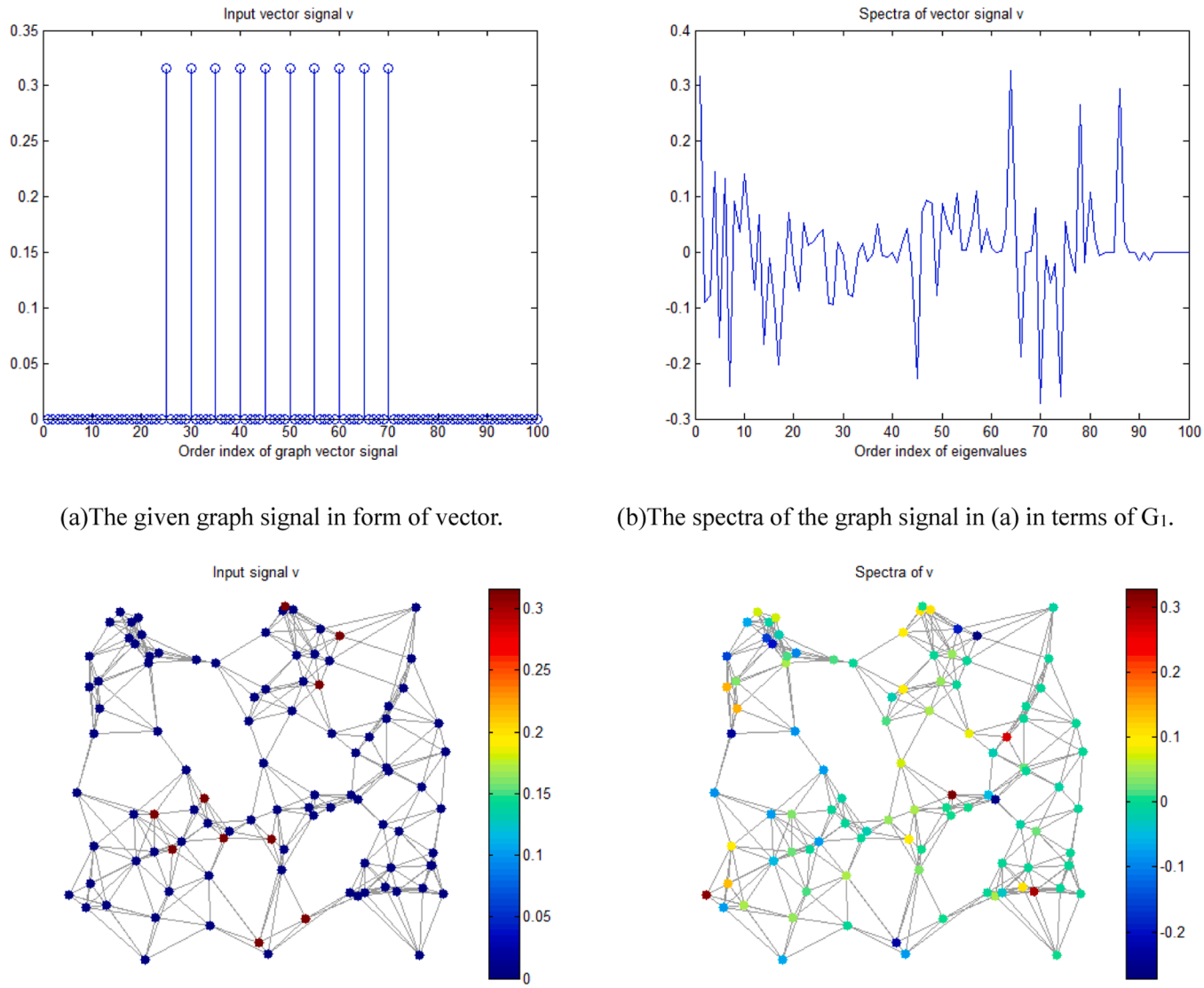
Proof. the proof is trivial because the relation can be found easily.

In the same manner, we could obtain the following lemma.

Lemma 5. *For a given graph* $\mathbf{G}_1$ *with its matrix signal* $\mathbf{V} = \sum_k\lambda_k\mathbf{v}_k\mathbf{v}_k^{\dagger}$ $(\lambda_k > 0)$ *with* $\mathrm{tr}(\mathbf{V}^2) < 1$ *and* $\mathrm{tr}(\mathbf{V}) = 1$, *if we have three series of standard orthogonal basis* $\mathbf{X} = [\mathbf{x}_1\ \mathbf{x}_2 \ldots \mathbf{x}_n]$, $\mathbf{Y} = [\mathbf{y}_1\ \mathbf{y}_2 \ldots \mathbf{y}_n]$ *and* $\mathbf{Z}_1 = [\mathbf{z}_{1,1}\ \mathbf{z}_{1,2} \cdots \mathbf{z}_{1,n}]$, $\mathbf{Z}_2 = [\mathbf{z}_{2,1}\ \mathbf{z}_{2,2} \cdots \mathbf{z}_{2,n}]$, *and two unitary transform matrix* $\mathbf{U}_1, \mathbf{U}_2$ *with* $\mathbf{Y} = \mathbf{U}_1\mathbf{X}$ *and* $\mathbf{X} = \mathbf{U}_2\mathbf{Z}$, $\mathbf{X}$ *is the graph Fourier basis of the graph* $\mathbf{G}_1$, *and* $\mathbf{Z}_1, \mathbf{Z}_2$ *is the reference Fourier basis of graph* $\mathbf{G}_2, \mathbf{G}_3$, *respectively, then we have the following uncertainty relation*

$$\min\left\{\mathbf{H}^{\mathbf{X}\|\mathbf{Z}_1}(\mathbf{V}), \mathbf{H}^{\mathbf{X}}(\mathbf{V})\right\} + \min\left\{\mathbf{H}^{\mathbf{Y}\|\mathbf{Z}_2}(\mathbf{V}), \mathbf{H}^{\mathbf{Y}}(\mathbf{V})\right\}$$
$$\geq \min\left\{\ln\frac{1}{\xi_1\cdot\xi_2}, \max\left\{0, \log\left\{\left(|\varsigma_1|^2|\varsigma_2|^2\right)^{nL\lambda_{\min}}\left(\frac{1}{|\zeta|^2}\right)^{nL\lambda_{\min}\left(|\xi_1|^2+|\xi_2|^2\right)}\right\}\right\}\right\}, \tag{16}$$

where the relative entropy $\mathbf{H}^{\mathbf{X}\|\mathbf{Z}_1}(\mathbf{V}) = \sum_{k=1}^{\mathrm{K}}\left(\lambda_k\sum_{i-1}^n|\mathbf{x}_i^{\dagger}\mathbf{v}_k|^2\log\frac{|\mathbf{x}_i^{\dagger}\mathbf{v}_k|^2}{|\mathbf{z}_{1,i}^{\dagger}\mathbf{v}_k|^2}\right)$, $\mathbf{H}^{\mathbf{Y}\|\mathbf{Z}_2}(\mathbf{V}) = \sum_{k=1}^{\mathrm{K}}\left(\lambda_k\sum_{i-1}^n|\mathbf{y}_i^{\dagger}\mathbf{v}_k|^2\log\frac{|\mathbf{y}_i^{\dagger}\mathbf{v}_k|^2}{|\mathbf{z}_{2,i}^{\dagger}\mathbf{v}_k|^2}\right)$, and $\zeta = \max_{i,j,k}\left\{|\mathbf{z}_{1,i}^{\dagger}\mathbf{v}_k|,\right.$



(a)The given graph signal in form of vector.

(b)The spectra of the graph signal in (a) in terms of G_1.

(c)Display of graph signal of (a) in graph G_1.

(d) Display of spectra in graph G_1.

Fig. 3. The graph signals and graphs in the first experiment.

$|\mathbf{z}_{2,j}^{\dagger}\mathbf{v}_k|\}$, $\xi_1 = \max_{i,k}(|\mathbf{x}_i^{\dagger}\mathbf{v}_k|)$ and $\xi_2 = \max_{i,k}(|\mathbf{y}_i^{\dagger}\mathbf{v}_k|)$, $\varsigma_1 = \min_{i,k}(|\mathbf{x}_i^{\dagger}\mathbf{v}_k|) > 0$ and $\varsigma_2 = \min_{i,k}(|\mathbf{y}_i^{\dagger}\mathbf{v}_k|) > 0$.

Proof. the proof is trivial and is omitted.

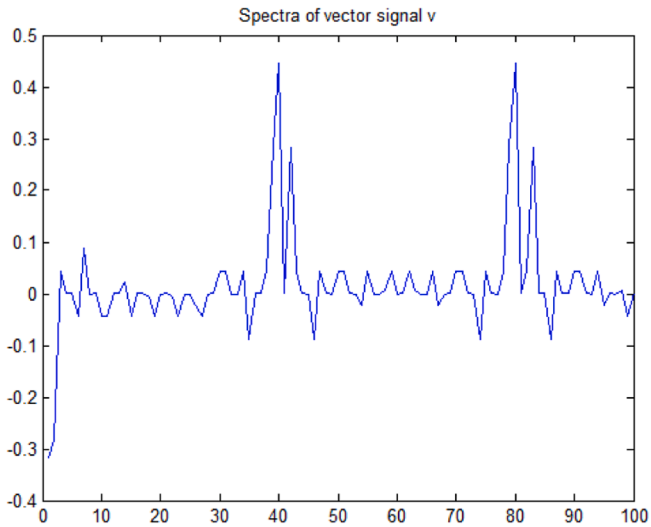
In other words, the two lemmas avoid the case of infinitely great quantity. For example, if the given reference basis or the given reference signal energy distribution is not appropriate and fully different from that of the client signal, we still can obtain the limited values of entropies to avoid the infinitely great quantity.

6. Numerical experiments and comparison

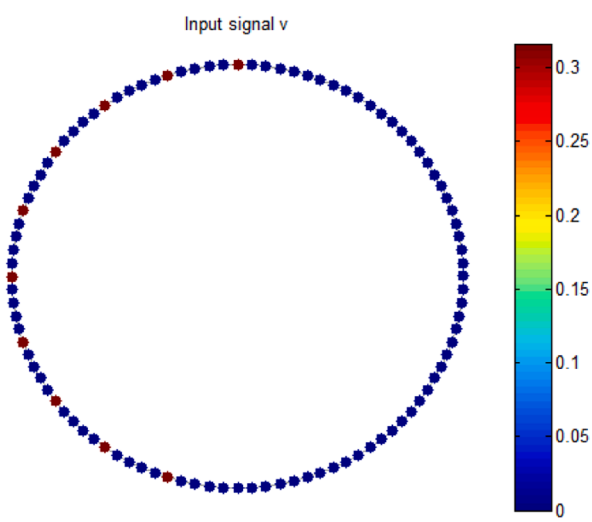
In the following section, we will show some numerical experiments on graph signals to demonstrate the efficiency of our proposed uncertainty relations.

Here we will show one numerical experiment to demonstrate the efficiency of Theorem 1. Here we let $\mathbf{X}$ be the graph Fourier basis of the graph $\mathbf{G}_1$ shown in Fig. 3(c and d), and $\mathbf{Y}$ is unit matrix, i.e., $\mathbf{Y} = \mathbf{I}$, $\mathbf{Z}_1 (= \mathbf{Z}_2)$is the reference Fourier basis of another graph $\mathbf{G}_2$ shown in Fig. 3(f and g). The graph vector signal $\mathbf{v}$ is shown in Fig. 3(a), more details can be found in Fig. 3.

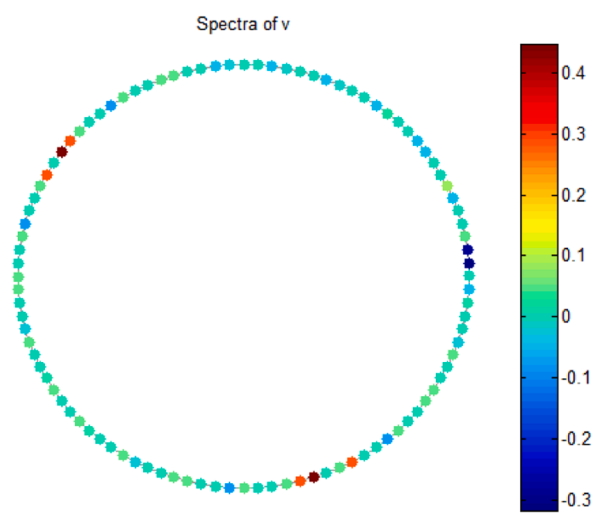
In this experiment, we have $\mathbf{H}^{\mathbf{X}\|\mathbf{Z}_1}(\mathbf{v}) \approx 2153$,$\mathbf{H}^{\mathbf{Y}\|\mathbf{Z}_2}(\mathbf{v}) \approx 43.1101$, and $\zeta \approx 0.4472$, $\xi_1 \approx 0.3280$, $\xi_2 \approx 0.3162$, $\varsigma_1 \approx 0.0001$, $\varsigma_2 \approx 0.3162$,$n = 100$, thus we have $\log\left\{\left(|\varsigma_1|^2|\varsigma_2|^2\right)^n \left(\frac{1}{|\zeta|^2}\right)^{n\left(|\xi_1|^2+|\xi_2|^2\right)}\right\} \approx -1613$, so we obtain the following relation



(e)The spectra of the graph signal in (a) in terms of G_2.



(f)Display of graph signal in (a) in graph G_2.



(g) Display of spectra in graph G_2.

Fig. 3. (*continued*).

$$2196.1101 \approx \mathbf{H}^{\mathbf{X}\|\mathbf{Z}_1}(\mathbf{v}) + \mathbf{H}^{\mathbf{Y}\|\mathbf{Z}_2}(\mathbf{v}) \geq \max\left\{\log\left\{\left(|\varsigma_1|^2|\varsigma_2|^2\right)^n\left(\frac{1}{|\zeta|^2}\right)^{n\left(|\xi_1|^2+|\xi_2|^2\right)}\right\},\quad 0\right\} = 0,$$

then Theorem 1 is verified. Here we note that in practice we know that the practical data are limited by the truncation error, we let our data to keep four decimal places during calculation.

Especially, if we set $\mathbf{Z}_1 = \mathbf{X}, \mathbf{Y} = \mathbf{Z}_2$, i.e., the reference Fourier basis are the same as that of the client graph $\mathbf{G}_1$, then we will have

$$0 = \mathbf{H}^{\mathbf{X}\|\mathbf{Z}_1}(\mathbf{v}) + \mathbf{H}^{\mathbf{Y}\|\mathbf{Z}_2}(\mathbf{v}) \geq \max\left\{\log\left\{\left(|\varsigma_1|^2|\varsigma_2|^2\right)^n\left(\frac{1}{|\zeta|^2}\right)^{n\left(|\xi_1|^2+|\xi_2|^2\right)}\right\}, 0\right\} = 0,$$

then Theorem 1 is verified as well under this special case.

Some similar numerical experiments on other theorems have been performed and the same conclusion has been obtained by us as well.

That is to say, if the given expectant (assumed) signal concentration and the practical signal concentration are the same, then the relative entropy will be zero, at this time, the uncertainty would be zero. In other words, there will be no uncertainty any more, which will give us great potential importance for our signal representation and analysis. Of course, how to find the optimal reference basis will be our future important work in our other study.

Instead, if the two given reference basis are the same (or the two given reference distributions are the same), then the uncertainty bound cannot reduce to zero, that is to say, after giving a reference basis including corresponding signal energy distribution, then any signal in two transform domains cannot have the same distribution any more as that of the given basis, which is the new extension of traditional uncertainty relations without relative entropy. Therefore, relative entropy based uncertainty principle would give us the new prospect for signal concentration analysis and time-frequency resolution analysis, and the new relations would support one new manner for uncertainty

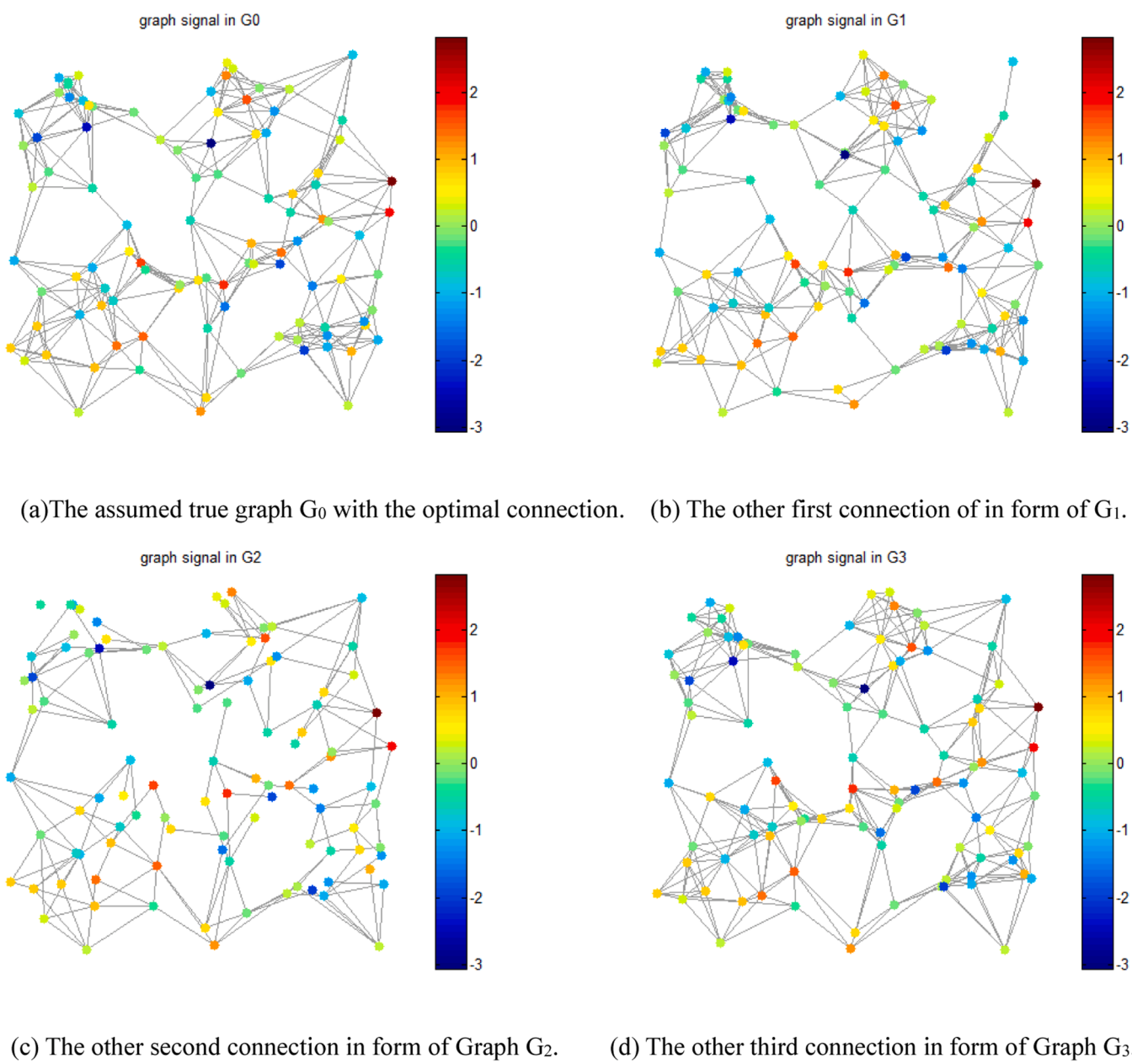
measurement.

7. Applications and analysis

In this section, we will show some applications of our new uncertainty relations on graph signals along with analysis to demonstrate the efficiency of our proposed uncertainty principles. Before beginning, we must note that the application of uncertainty principle in signal processing lies in two aspects: one is time-frequency resolution analysis [4-17] in different transform domains, and the other one is signal sparse representation [19-36]. Of course, GSP belongs to signal processing and it would have the two applications as well. On the other hand, since the relative entropy measures the distinguishability between the known (priori) basis and the client basis, generally the lower bounds would have high resolution, as is trivial and we won't pay more attentions. Therefore, like these papers [19-36], more attentions would be paid on graph signal sparse representation. Specially, an important task in signal sparse representation is to find the optimal representation bases to represent the client signal so that we could obtain the sparest representation [34-39]. In other words, the optimal graph signal sparse representation is indeed to find the most similar counterpart (i.e., the same counterpart in the best case) that is existing or given in advance, or to say, to find the most similar reference bases.

In weather forecast and analysis, the historical similarity weather analysis is an important manner, and the employment of historical similarity weather analysis is very important for the understanding and weather forecast of weather system [53]. On the other hand, meteorological observation is used nearly everywhere in practice. All these applications need the typical graph signals being lots of irregular graph data in essence (i.e., the meteorological elements from all kinds of meteorological sensors) such as temperature, humidity, atmospheric pressure and wind and so forth. However, for the same network system of meteorological sensors, different people will give or use different topological structures to process these data as shown in Fig. 4.

In Fig. 4 (note the different topological structures), we assume we have a network system of meteorological sensors in certain city or place.



(a)The assumed true graph G_0 with the optimal connection. (b) The other first connection of in form of G_1.

(c) The other second connection in form of Graph G_2. (d) The other third connection in form of Graph G_3.

Fig. 4. One simulated meteorological sensor network of the same city by different people.

At the same time, we have three groups to process these graph data. We find that they will use different topological structures because the connection is variable and it is hard to say which connection must be the optimal. Here we assume that the first graph G_0 in Fig. 4(a) is our ideal (the optimal) connection. Now we want to know which one between G_1-G_3 from Fig. 4(b) to Fig. 4(d) is the closest to the ideal one in Fig. 4(a). Thus we could use our proposed relative entropy based uncertainty relations to judge which graph is the optimal between G_1-G_3 from Fig. 4(b) to Fig. 4(d) based on the first graph G_0 in Fig. 4(a).

We assume we have one time's graph data (without loss of generality, we use the random Gaussian white noise to denote our data) shown in Fig. 4. Form Theorem 1, now we can obtain these relative entropy based uncertainty values and bound values as follows.

At first, we let G_0 be our client graph and G_1-G_3 be our reference graphs, respectively.

For G_0 and G_1, we have $\mathbf{Z}_1 = \mathbf{X} = \mathbf{I}$, and $\mathbf{Y}$is the graph Fourier base from G_0, $\mathbf{Z}_2$ is the graph Fourier base from G_1 . Thus we have

$$\begin{aligned} 1.506 = 0 + 1.5064 &= \mathbf{H}^{\mathbf{X}\|\mathbf{Z}_1}(\mathbf{v}) + \mathbf{H}^{\mathbf{Y}\|\mathbf{Z}_2}(\mathbf{v}) \\ &\geq \max\left\{\log\left\{\left(|\varsigma_1|^2|\varsigma_2|^2\right)^n\left(\frac{1}{|\zeta|^2}\right)^{n\left(|\xi_1|^2+|\xi_2|^2\right)}\right\}, 0\right\} = \{-0.3291, 0\} \\ &= 0. \end{aligned} \tag{17}$$

Similarly, for G_0 and G_2, we have $\mathbf{Z}_1 = \mathbf{X} = \mathbf{I}$, and $\mathbf{Y}$ is the graph Fourier base from G_0, $\mathbf{Z}_2$ is the graph Fourier base from G_2 . Thus we will get

$$\begin{aligned} 303.5370 = 0 + 303.5370 &= \mathbf{H}^{\mathbf{X}\|\mathbf{Z}_1}(\mathbf{v}) + \mathbf{H}^{\mathbf{Y}\|\mathbf{Z}_2}(\mathbf{v}) \\ &\geq \max\left\{\log\left\{\left(|\varsigma_1|^2|\varsigma_2|^2\right)^n\left(\frac{1}{|\zeta|^2}\right)^{n\left(|\xi_1|^2+|\xi_2|^2\right)}\right\}, 0\right\} \\ &= \max\{16.0987, 0\} = 16.0987. \end{aligned} \tag{18}$$

At the same time, for G_0 and G_3, we have $\mathbf{Z}_1 = \mathbf{X} = \mathbf{I}$, and $\mathbf{Y}$ is the graph Fourier base from G_0, $\mathbf{Z}_2$ is the graph Fourier base from G_3 . Thus we will obtain

$$\begin{aligned} 110.8493 = 0 + 110.8493 &= \mathbf{H}^{\mathbf{X}\|\mathbf{Z}_1}(\mathbf{v}) + \mathbf{H}^{\mathbf{Y}\|\mathbf{Z}_2}(\mathbf{v}) \\ &\geq \max\left\{\log\left\{\left(|\varsigma_1|^2|\varsigma_2|^2\right)^n\left(\frac{1}{|\zeta|^2}\right)^{n\left(|\xi_1|^2+|\xi_2|^2\right)}\right\}, 0\right\} \\ &= \max\{7.0954, 0\} = 7.0954. \end{aligned} \tag{19}$$

Clearly, we find that the graph G_1 in Fig. 4(b) is the closest to the ideal one in Fig. 4(a). In other words, graph G_1 is the optimal between G_1-G_3 from Fig. 4(b) to Fig. 4(d) from the sense that is the closest to the ideal one via our uncertainty relations. At the same time, we find there is no need to calculate all the relative entropic values shown in the left of Eqs. (17)-(19). We only need to calculate the bounds shown in the right of Eqs. (17)-(19) and we also can obtain the same result: the graph G_1 in Fig. 4(b) is the optimal because it has the smallest bound value among these three ones. Moreover, we have replaced the given noise signal with multiple different random Gaussian white noise and the results are always similar and the conclusion is the same as well so that we only show certain time's result here.

Of course, this example is only one kind of simple application belonging to the signal sparse representation. We must note that the application of uncertainty principle in time-frequency resolution analysis [4-17] and signal sparse representation [19-39] maybe give more theoretical hints instead of direct usage. In the similar manner, there are other practical applications such as video network monitoring, urban road congestion monitoring and security monitoring and so on. For the sake of simplification of this paper, here we only show the network system of meteorological sensors.

8. Conclusions

In this paper, inspired by the physical quantum mechanics in terms of uncertainty principle in presence of quantum memory, some novel relative entropy based uncertainty relations on graph signals have been addressed carefully. The main contributions can be concluded as follows.

First, the relative entropy based uncertainty principle in terms of graph vector signal is derived for the first time and it shows the relation between the new bound and the parameters: the length n of the vector $\mathbf{v}$, the min non-zero correlation coefficients ς_1 and ς_2, and the max correlation coefficients ζ, ξ_1 and ξ_2. At the same time, the special cases are involved as well.

Second, the relative entropy based uncertainty principle in terms of graph matrix signal is derived for the first time as well and it shows the relation between the new bound and the parameters: the length n of the vector $\mathbf{v}$ in matrix eigendecomposition, the number of eigenvalues of matrix: K, the min non-zero correlation coefficients ς_1 and ς_2, and the max correlation coefficients ζ, ξ_1 and ξ_2, the min non-zero eigenvalue $\lambda_{\min}$. In addition, some special cases are shown as well.

Third, in order to avoid the infinitely great quantity, two new combined uncertainty relations are demonstrated at the same time according to the possible case that the given reference signal's energy distribution is not appropriate and fully different from that of the client signal, we still can obtain the limited values of entropy.

Finally, some numerical experiments and certain application are given in great details including theoretical analysis to show the efficiency of these proposed uncertainty relations.

In one word, these novel uncertainty relationships with lower bounds give us the possible advantages over classical counterpart. That is to say, if the given expected signal concentration (energy distribution) and the practical signal concentration are the same, then the relative entropy will be zero, the uncertainty would be zero. In other words, there will be no uncertainty any more, which will give us great potential importance for our signal representation and analysis. Instead, if the two given reference distributions are the same, then the uncertainty bound cannot reduce to zero, which is the new extension of traditional uncertainty relations.

Therefore, the newly proposed relative entropy based uncertainty principle would give us the new prospect for signal concentration (e.g., for signal sparse representation) and time-frequency resolution analysis, and the new relations would yield one new way for uncertainty measurement in future.

CRediT authorship contribution statement

Xu Guanlei: Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Data curation, Conceptualization. **Xu Xiaogang:** Project administration, Investigation, Funding acquisition. **Wang Xiaotong:** Funding acquisition, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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